

Infrastructure as Market Design: Location and Technology Choice on the Electric Grid*

Drew Van Kuiken

September 17, 2026

Abstract

Economists have long recognized that infrastructure is important for welfare, but figuring out the marginal benefit of infrastructure is difficult. In this paper, I study how infrastructure shifts the equilibrium level, location, and technological composition of entry. My setting is the Texas electric grid, where transmission infrastructure is scarce and production varies by location and technology. I solve a dynamic entry game for renewable generators, capturing spatial and technological entry margins. Embedded within my entry game is a novel, physically realistic model of electricity dispatch, where electricity flows and prices are endogenous. The model is estimated using data from 2018-2024 with generation, transmission capacity, and demand information. I show that infrastructure provision policies distort entry locations and technology choices when generators and grid operators do not jointly solve for entry and infrastructure provision. I explore two counterfactuals – one where entrants pay for their own grid upgrades, and one where grid operators create a market for new transmission – under fixed demand and data-center-driven demand growth. I find that the market for new transmission generates more welfare than the Texas or entrant-funded upgrade policies, and that both non-market policies yield inefficiently low and spatially distorted entry choices.

*I am grateful to Jon Williams, Andy Yates, Gary Biglaiser, Marco Duarte, and Valentin Verdier for their help. I thank seminar and conference participants for helpful comments on this paper. Department of Economics, University of North Carolina at Chapel Hill, drew.van.kuiken@unc.edu.

Economists have long recognized that infrastructure is important for welfare, but figuring out the marginal benefit of infrastructure is difficult (Fernald, 1999; Duranton and Turner, 2011; Brancaccio et al., 2024). One potential reason: infrastructure availability shifts firm entry decisions. Prior literature suggests that the spatial allocation of firms depends on transport costs (e.g., Krugman 1991 Increasing Returns and Economic Geography, Desmet and Rossi-Hansberg 2014) and rival firm positioning (e.g., (Seim, 2006), (Jia, 2008)). This paper begins by observing that if benefits to infrastructure vary by firm, shifting infrastructure availability will change the equilibrium level and location of entry. Thus welfare changes depend on how infrastructure shifts the composition of firm entry, particularly when firms differ in their productivity, costs, or consumers they serve.

In this paper, I study how infrastructure policy shifts equilibrium level, location, and technological composition of entry in the context of the Texas electric grid, where these dynamics are particularly relevant. Texas has ample wind and sunlight resources. It also builds new electric transmission infrequently. These two features of the market came to a head during the 2018 to 2024 period, when fixed costs for wind and solar generators dropped and both technologies entered rapidly. Grid capacity remained mostly fixed throughout the period and grid congestion increased. The incidence of this congestion was unequal: generators stuck behind congested transmission lines saw revenues fall, while generators in uncongested regions of the grid earned steady revenues. The geography of resource quality made this particularly punishing for wind generators – while solar generators had access to reasonably high quality sunlight throughout the state, high quality wind resources were located in the panhandle and central Texas, gated behind congested transmission elements. I show that these developments are consistent with a coordination inefficiency in infrastructure provision: limited transmission availability distorts entry decisions, and distorted entry decisions reduce the returns to investing in transmission.

To test the effect of infrastructure on entry, I solve a dynamic entry game among renewable generators, where generators differ in their technology and their location on the grid. Entrants are forward-looking and make decisions based on the expected evolution of entry both in their own region and in other regions. I find that limited transmission capacity interacts with wind’s lack of geographic flexibility to reduce wind entry, while solar generators sacrifice sunlight quality to pursue available transmission capacity.

Next, I ask whether alternative transmission policies would lead to welfare-improving allocations of generation on the grid. I begin by requiring entrants upgrade the grid when they enter, which targets the coordination inefficiency. The policy is based on a transmission policy used throughout the rest of the U.S. I find that this policy imposes inefficiently high costs on entrants, leading to cascading withdrawal and lower welfare than the Texas baseline.

I then test the effect of setting up a market where entrants, incumbents, and consumers reveal their willingness to pay for new transmission. I find this policy lowers congestion, lowers prices, increases entry in high resource quality regions, lowers carbon emissions, and constructs more transmission than Texas’ baseline policy, resulting in a XXX% increase in welfare. Finally, I consider how infrastructure policies deal with data-center-driven demand growth, adding 1 GW of demand in different regions on the grid. I find that the market for transmission greatly outperforms existing policies in this case, increasing welfare by YYY% over the Texas baseline.

Together, my results suggest three takeaways: (i) infrastructure is non-neutral towards differentiated technologies, shaping firms’ equilibrium locations and technology choices on the electric grid; (ii) without accounting for endogenous entry, transmission policy builds inefficiently little transmission and equilibrium entry decisions are distorted spatially and technologically; (iii) coming data center demand is likely to amplify existing entry inefficiencies, but better infrastructure policy can leverage increased demand to build more transmission and lower electricity prices.

The analysis is built on a newly assembled dataset, featuring detailed information on generation, transmission capacity, and demand for the Texas electric grid. I show how these objects interact to determine prices in a simplified model of the grid. I use the simplified model to illustrate how the model of electricity dispatch works, generating endogenous electricity flows, prices, and quantities sold. The model captures the physical realities that generators must contend with: when transmission is congested, generators in counties upstream of the congestion cannot send their electricity to the highest demand regions. Prices in these counties fall as a result, often reaching levels near zero. Prices in counties downstream of the congestion rise. I then discuss a key inefficiency that emerges from the simplified model: when entrants and grid operators cannot coordinate on infrastructure provision, entry incentives are spatially and technologically distorted. These results hold for congestible infrastructure broadly.

The entry model has 40 regions, which span the service territory of the Electric Reliability Council of Texas (ERCOT), the grid operator for Texas. Each period, a fixed number of potential entrants arrive in each region and decide on entry. Regions vary in the quality of resource they have available and the expected price that entrants will earn from producing in that region, among other factors. Resource quality is exogenous and fixed; regional prices and quantities are endogenous. Entrants make decisions based on expected profits, weighed against the fixed costs of entry for a given region and a cost shock. I treat ERCOT’s infrastructure policy as a fixed grid policy with no transmission capacity improvements,

consistent with the level of improvements ERCOT made to the grid over this period.¹

Prices on the electric grid depend on the flows of electricity throughout the network. Solving the entry game while accounting for evolving wind and solar generation in all 40 regions would require a computationally infeasible state space. Instead of solving the full game, I solve a simplified game, where entrants form beliefs about other-region entry based on equilibrium entry policies and integrate over potential spatial allocations of generation that cost shocks could justify. This preserves the network structure of prices: entry in one region affects prices everywhere. In equilibrium, entrant beliefs align with the stationary distribution of generation allocations, meaning that entrant expectations about prices are accurate within the model.

The entry model confirms that wind generators are more heavily affected by congestion than solar generators: in a partial equilibrium exercise, I find that setting congestion costs equal to zero increases wind entry by 8.9%, primarily in the regions where wind generation already exists, while solar entry increases by less than 2%.² Testing whether alternative infrastructure policies lead to efficient entry and increased welfare remains.

The first counterfactual implements a policy that requires entrants upgrade the grid themselves, which I refer to as the upgrade-on-entry policy. I find that upgrade-on-entry elicits demand for new transmission, but imposes costs of entry that are too high and borne exclusively by entrants when new transmission also benefits incumbents. Grid operators that used this policy generally share upgrades costs across entrants. Because upgrade costs are shared within entry cohorts, the consequences of adopting this policy are significant: entrants often withdraw instead of paying their upgrade cost allocation, leaving upgrade costs split among fewer entrants. In some cases, withdrawal decisions cascade. This reduces welfare by XXX% relative to the Texas baseline.

The second counterfactual implements a policy where entrants, incumbents, and consumers submit bids for new infrastructure, which I refer to as the market for transmission. I find that the market for transmission builds more infrastructure than either the baseline or upgrade-on-entry policies. The market gives the grid operator a way to elicit entrant, incumbent, and consumer values for new transmission. Increasing transmission availability reduces the entry distortions across technologies and locations, leading to increased entry across technologies and a spatial reallocation of solar entry towards high-resource quality

¹ERCOT frequently makes reliability focused upgrades to the grid, which are initiated based on concerns about line overloading and more commonly targeted at the lower-voltage distribution grid. On rare occasions, ERCOT has made substantial upgrades to its high-voltage transmission grid. A significant upgrade was made in 2014; more recently, proposed 765 kV transmission lines have been met with stiff political and popular opposition. I abstract away from the more political processes that govern these decisions.

²The spatial allocation of solar instead shifts, as generators enter in higher quality sunlight regions.

regions. The market for transmission improves welfare by XXX% over the Texas baseline.

In the near term, demand from data centers is poised to cause new challenges for the grid. Whereas electricity demand has been mostly flat over the past 25 years, Texas grid operators expect demand to grow by XXX% over the next YYY years. In a second counterfactual, I examine how the Texas baseline, the upgrade-on-entry policy, and the market for new transmission all respond to the addition of a 1 GW data center to the grid. I find....

Literature Review

Past studies have shown that policy shifts what products firms offer (Sweeting (2013)) and that the welfare effects of changing product positioning are important (Wollmann (2018)). This literature studies how firms choose the products they sell and the locations they serve. I connect this to a group of papers that study how infrastructure affects welfare (Fernald (1999); Duranton and Turner (2011); Brancaccio et al. (2024)). The electric grid is uniquely well suited to understand endogenous technology choice and siting decisions: wind and solar production profiles depend on exogenous environmental factors and locations are differentiated based on shared transmission infrastructure. I leverage this setting to estimate a dynamic entry model with endogenous congestion and transmission provision. The model captures how infrastructure policy changes the equilibrium assortment of wind and solar generation. I show that geographically flexible solar projects can move towards demand centers when infrastructure is scarce, while wind projects are constrained by the environment, resulting in under-entry. I am additionally able to use this model to consider how alternative grid management policies would shift the spatial allocation of firms, the equilibrium technology mix, and welfare.

This paper adds to a growing literature concerned with the electric grid. Grid management is complicated and economists have only recently begun to model the processes that RTOs use to govern entry and competition on the grid (Wolak (2015); Gowrisankaran et al. (2016); Cicala (2022); Mays (2023); Johnston et al. (2024); Boatwright (2025); Cannière (2026)). I add to these papers the first embedding of a realistic dispatch model within a dynamic entry game. I use data from existing generators and the offers they submitted to the grid operator, a novel representation of the grid, and endogenize congestion in a way that is new to the literature. Previous work has used representations of the grid (Arkolakis and Walsh (2023); Golden (2026)), but focused on the grid as a series of bipartite connections. This paper instead models the physics that govern where power flows on the grid. This allows me to model how congestion on one grid segment affects prices throughout the grid. More broadly, it allows me to credibly estimate the dynamic incentives firms face when choosing where they should enter the grid and consider how different grid management policies would

shift the level and location of renewable entry.

The benefits and underprovision of transmission has been studied in a variety of settings, including the Midwest, Texas, Chile, and the continental U.S. (Borenstein et al. (2000); Fell et al. (2021); Gonzales et al. (2023); Arkolakis and Walsh (2023); Hausman (2024); Holland et al. (2024); Boatwright (2025); Golden (2026)). These papers show that transmission increases allocative efficiency, decreases emissions, and increases long-run investment in renewable capacity. Despite widespread agreement as to the value of new transmission, underinvestment in transmission persists. On this point, my paper is closest to Boatwright (2025), who also studies entry to the grid in Texas. She estimates a two-stage model of developer applications and project completion, showing how observed congestion exposure affects entry into the interconnection queue and eventual construction. In contrast, I model the dynamic feedback between entry, dispatch, and congestion: firms' siting decisions change future grid conditions, changing the value of entry. In counterfactuals, I add in transmission planning, which increases the dynamic feedback because grid upgrades respond to the state of the network. By endogenizing the transmission planning process, I connect this literature to an older literature that theorized about transmission planning policies for RTOs (Léautier (2000); Vogelsang (2001); Joskow and Tirole (2005); Hogan et al. (2010)). These papers recognize the difficulty in incentivizing efficient expansion of the transmission grid, showing how underinvestment in transmission arises. I capture this underinvestment using my model of centralized transmission planning. My main contribution to these literatures is to estimate the costs of centralized transmission planning when entry is endogenous. I also propose a new mechanism of transmission funding that accounts for the strategic entry of renewable energy and which leads to larger investment in the grid.

The rest of the paper is structured as follows: Section 1 introduces the setting. Section 3 introduces the dynamic entry model, while Section 4.2 describes how I estimate the model. Section 5 describes the entry model results and Section 6 introduces the counterfactuals and the results I find. Section 7 concludes.

1 Background

The Texas electric grid, operated by ERCOT, is an independent power grid, meaning that the vast majority of electric generation demanded by Texas consumers is met by Texas-based generators (Cohn, 2022).³ As system operator, ERCOT is responsible for balancing supply and demand for electricity on the grid, which it accomplishes by scheduling when and how

³Texas has two transmission lines connecting it to the Eastern Interconnection and two transmission lines connecting it to Mexico; these lines are primarily used for scheduled and emergency power trades.

much electricity individual generators should produce. There are more than 1,460 generators in ERCOT’s territory (Electric Reliability Council of Texas, 2026).⁴ Electricity in ERCOT is transported using high-voltage alternating current transmission lines that link the resource-rich, low-demand regions in the western part of the state to high-demand population centers in the east.

In the short-run, demand for electricity is highly inelastic. Every five minutes, demand for electricity (referred to as load) is reported by utilities and wholesale consumers. Generators themselves submit bid curves to the grid operator, reporting the amount of electricity they are willing to provide at different prices.⁵ Balancing supply and demand then requires the grid operator solve a constrained optimization problem: given demand for electricity, find the least-cost dispatch of electricity generation subject to transmission capacity. The operator solves this problem for each node across the entire grid. Nodes are the point at which electricity moves from generators to the transmission grid and the point at which demand is settled for a small area. There are several thousand nodes in ERCOT.

Constraints in the optimization problem include transmission line capacity. Once lines become congested, the grid operator is forced to dispatch electricity from nodes with higher cost than the system cost and curtail electricity production from nodes with lower cost than the system cost. In nodes where generators are curtailed, the local price (LMP, or locational marginal price) falls below the system price of energy. LMPs indicate what the marginal generator in a node is willing to produce electricity for. In nodes on the demand side of congestion, prices will rise above the system price of energy, as marginal generation in that node has to turn on to meet demand.

ERCOT is responsible for setting generator entry policy. ERCOT uses a unique policy among American grid operators, where generators connect freely to the grid, whether or not the transmission capacity exists for them, and take on the risk of curtailment as a consequence (Trabish, 2025).⁶ Curtailment is when a generator has to throw away the electricity it produces because there is insufficient transmission capacity to carry the

⁴Moving forward, I refer to ERCOT’s territory as ERCOT for simplicity.

⁵I assume generators report costs honestly. Independent market monitors are appointed to each RTO to ensure that reports are honest. Market monitors are able to access internal generator data and grid operator data to ensure that generators are not manipulating their offers.

⁶Most other grids use an “invest-and-connect” process. When a generator enters the grid, the operator initiates a study process to identify any overloads that the generator might cause and suggests a series of improvements to the grid that the generator should make to avoid grid congestion in summer peak, winter peak, and light demand scenarios. The generator must then make investments to improve the grid before it can enter. The scope of the upgrades that a generator must make thus depends on the location it chooses: locations with poorer local transmission capacity are more likely to trigger expensive upgrades. The grid is public infrastructure and investments are non-rival up to grid capacity. I return to the invest-and-connect policy in counterfactuals.

generator’s electricity to demand.

Grid upgrades are centrally planned in ERCOT. Generally, utilities are responsible for identifying and proposing new upgrades, while grid operators retain final approval authority. Utilities earn a regulated return on these grid investments, with rates subject to review by Public Utility Commissions. Recent studies have found that transmission planning is falling far behind generation growth and, more recently, load growth (Congressional Budget Office, 2026). Today, 90% of upgrades are focused on reliability (i.e., avoiding transmission line overloads), instead of economic benefits (i.e., reducing congestion costs). On one occasion, legislators in Texas, along with ERCOT and utilities, decided to fund larger transmission projects explicitly to facilitate the development of wind generation – the CREZ lines in 2014 (Jankovska and Cohn, 2020). There has not been a significant new transmission project in ERCOT since the CREZ lines.

The entry of renewables in ERCOT between 2018 and 2024 led to unprecedented congestion, with yearly congestion costs reaching a high of \$2.8B in 2022 (Shreve et al. (2025)). Transmission upgrades have not kept pace with the renewable capacity and transmission availability has decreased as a result. Table 1 shows the yearly congestion costs and transmission line miles upgraded in ERCOT (where upgrades represent increased transmission capacity). The table subsets transmission line upgrades to 230 kV and higher capacity lines, of which there are 18,446 miles on the grid. Across the seven year period, roughly 5% of transmission lines received an upgrade.

Table 1: Transmission Congestion Costs and Transmission Line Upgrades

	2018	2019	2020	2021	2022	2023	2024
Congestion costs (\$ millions)	1,260	1,260	1,400	2,100	2,800	2,400	1,900
Transmission Line Upgrades (miles)	292	0	0	373	0	213	121

Notes: Congestion costs are in millions of dollars and come from Table B1 of Shreve et al. (2025). Transmission Line Upgrades measures completed new 230 and 345 kV circuits, including new corridors and additional circuits on existing corridors. Rebuilt lines, station work, and generator interconnection facilities are excluded.

The technology mix of generation in ERCOT has changed dramatically in recent years. Figure 1(a) shows generators that operated in ERCOT in 2000. Generators clustered around major urban areas and industrial load zones like the Gulf Coast. Electric generation mostly came from dispatchable resources like coal and natural gas. Dispatchable resources are significant in that the amount of electricity they generate is a choice - if the grid needs more electricity, dispatchable resources can increase their production until they reach their high sustained limit, which is a function of their nameplate capacity.

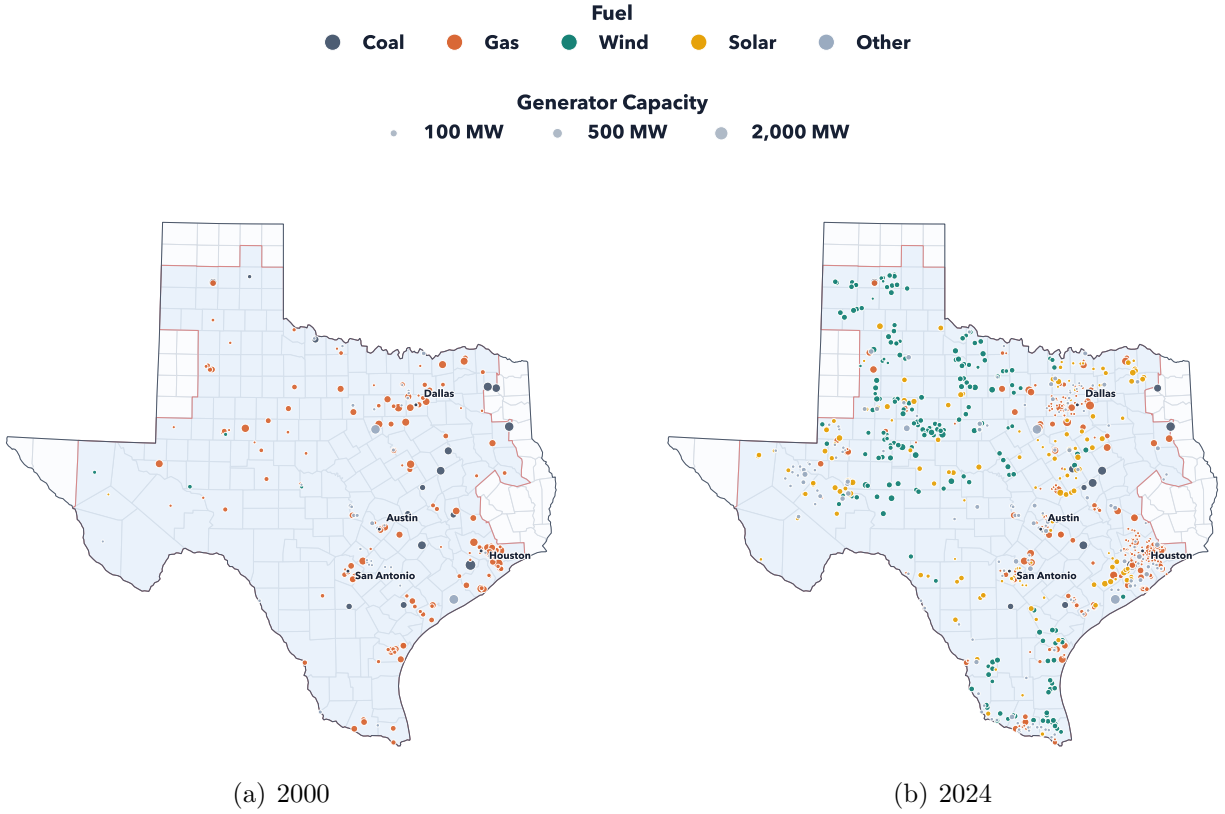


Figure 1: ERCOT generating capacity, 2000 and 2024

Notes: Each point represents a generator. Marker area scales with nameplate capacity, and color denotes the facility’s dominant fuel category. “Other” includes nuclear, hydroelectric, biomass, storage, and other fuels. Pale-blue shading denotes counties in ERCOT; the light-red line traces the resulting county-level ERCOT boundary. Data come from the historical EIA-860A/B inventories for 2000 and the final EIA-860 2024 release.

Figure 1(b) shows generators that operated in ERCOT in 2024. The rise of renewables in ERCOT has been remarkable. Whereas coal and gas represented 92% of capacity in 2000, they represent 53% in 2024. Renewables represent 38% (U.S. Energy Information Administration, 2026).⁷ The majority of renewable additions have come since 2017, following dramatic decreases in solar fixed costs and generous tax incentives for the development of wind power (U.S. Energy Information Administration, 2026).

ERCOT made changes to its grid to support renewables in 2014, when it finished building the CREZ transmission lines.⁸ The CREZ lines are high capacity transmission lines that link wind-rich regions like the panhandle and central Texas to demand centers in the eastern part of the state. The CREZ lines spurred development of wind generation in Texas (Jankovska and Cohn, 2020). Wind entrant locations shifted once they were built: pre-2014, the MW-weighted average distance between wind entrants and Texas’ major demand centers (Dallas-Fort Worth, Houston, Austin, and San Antonio) was 197 miles; in 2014, additions were located a MW-weighted average of 285 miles from demand centers.

Renewables continued to enter in the years following the CREZ lines. The MW-weighted average distance to demand centers for wind generators in 2018 was 208 miles. For solar, it was 212 miles. The congestion that arose during the 2018 through 2024 period generated very different equilibrium entry patterns for wind and solar generators. In 2024, wind generators located, on average, 189 miles from demand centers. Solar generators on the other hand located 148 miles away from demand centers, a 30% decrease in average distance relative to 2018, when congestion costs were lower.

2 Problem Intuition

In this section, I detail the model I use to recover electricity flows and the spatial and technological siting distortion it reveals.

Figure 2 shows a simplified three-node network of the grid. Load is located exclusively in the East node, with 120 MW demanded in total. There are three generators on the grid: a natural gas plant in the East node with constant marginal cost of \$30 / MWh and two wind generators in the West node with zero marginal cost.⁹ There are three transmission lines: a WE line, a WS line, and a SE line. I assume each line has a total capacity of 30 MW.

⁷Renewables produced 34% of energy consumed in ERCOT in 2024 because they only infrequently produce at full capacity (Electric Reliability Council of Texas, 2025).

⁸At the time of CREZ, solar capital costs were too high to be competitive as a utility electricity source in Texas, so the CREZ lines were primarily situated to facilitate wind generation.

⁹Gas generators typically has increasing marginal costs in the data. Using constant marginal costs here simplifies the analysis, at the cost of understating the inefficiency I describe below.

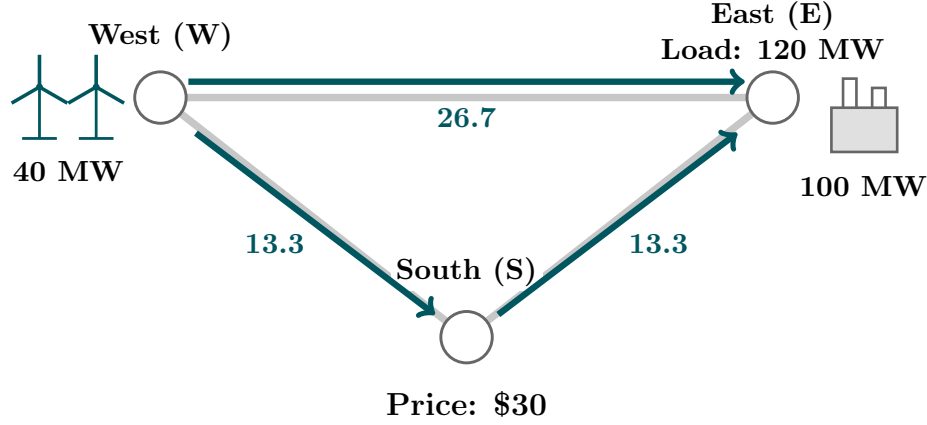


Figure 2: Illustrative three-node network

The numbers by each line indicate the amount of electricity that that line is carrying. Because the gas plant and load are in the same node, the gas plant does not need to use the transmission network. No transmission lines are congested, meaning all 40 MW of wind electricity can be delivered to load. The route that electricity travels to reach load depends on the reactance of the transmission lines. Reactance indicates how strongly a line limits power flows. I assume equal reactances across each line in this model. The volume of electricity that travels over a line is calculated as the relative share of the inverse of the sum of reactances. For example, the sum of reactances for line WE is 1, and for line WS-SE is 2. The share of electricity that travels on WE is thus $1/(1+1/2)=2/3$. $1/3$ travels on WS-SE. I calculate reactances for every transmission line in the grid that I use in the model. I describe how I do this in Appendix B.

The price of electricity depends on generator supply curves. When transmission lines are uncongested, supply curves are aggregated across nodes (Cannière (2026) refers to this as market integration). In this example, because wind has zero marginal cost, the market supply curve sits at $P = 0$ from 0 to 40 MW. Then, the gas plant becomes the marginal generator and prices rise due to gas' marginal cost. 80 MW of gas is used to meet 120 MW of load. The clearing price is \$30, which the gas and wind generators all earn. Prices are common across the grid when there is no congestion.

Next, consider a solar generator entering at node S. I assume that the solar generator produces 20 MW of generation at node S. Reactances dictate flows for the solar generator: $2/3$ of its 20 MW travel to load via line SE, while $1/3$ travels via SW-WE. This leads to an overload on line WE: $20/3 + 80/3 > 30$. To prevent the line from breaking, wind generation is curtailed and produces 35 MW instead of 40. The congestion breaks market integration. Prices depend on the marginal value of generation at each node. Because wind is curtailed,

marginal generation has no value, and the price at node W drops to \$0. Adding 1 MW of generation to node S has two effects: first, it delivers $2/3$ MW of electricity via line SE. Second, it adds $1/3$ MW of electricity to line WE. This again causes the line to overload, necessitating that wind generation at node W drop by $1/2$ MW. Since $1/3$ of the $1/2$ MW decrement travels via WS-SE and no new electricity travels via WE, the net delivery from a 1 MW injection at node S is $2/3 - (1/3 \times 1/2) = 1/2$. The price of electricity at node S is thus \$15, because it displaces $1/2$ MW of \$30 gas in node E. Finally, the price of electricity in node E stays at \$30, which is the marginal cost of gas generation.¹⁰

Most objects in the simplified example are available directly in the data. Transmission line geometry and voltage as of 2024 is available from the Homeland Infrastructure Foundation Level Data resource. Generator county information is available in ERCOT’s MORA reports. Generator supply curves are reported by ERCOT, using its SCED (Security Constrained Economic Dispatch) data. Load is available at the zonal level. Congestion data (including the substations between which congestion arose and the cost of that congestion for the grid) is available. I map congestion data to transmission line geometry by hand-encoding the location of relevant substations. Transmission line capacity is unobserved. I estimate capacity based on implied flows using generator supply information, congestion data, and the reactance assumptions. Together, these objects constitute what I call the dispatch model, the estimation of which is detailed in Section 3.3.1.

2.1 Siting Inefficiency

Consider the solar generator siting problem. Assume the generator can choose between entering in node S or in node W. Further, assume the generator produces 25 MW in node W and 20 MW in node S (due to higher resource quality). Let entry impose a fixed cost of 200 on the generator and upgrading line WE cost 250. Let the upgrade be large enough to remove the constraint under any siting choice. Table 2 summarizes the resulting profits and welfare.

¹⁰I formally derive the grid operator’s constrained optimization problem in Section XX.

Table 2: Solar siting and transmission-upgrade payoffs

Solar location	No WE upgrade		Upgrade WE	
	Solar profit	Welfare	Solar profit	Welfare
South (S)	100	250	400	150
West (W)	-200	-50	550	300

Entering S without an upgrade is the case analyzed above. Entering W without an upgrade worsens congestion on the grid, meaning the solar generator earns \$0 for its production. The solar generator provides 5 MW of production to the grid that previously did not exist, displacing 5 MW of gas generation for a benefit of 150. Because the solar generator pays 200 to enter, net welfare change is -50 .

Upgrading WE changes payoffs and welfare. With an upgrade, prices are equal at each node. Gas is still the marginal generator in each scenario, meaning prices are \$30 in all cases. If the solar generator enters S, it produces 20 MW. It earns $20 \times 30 - 200$. Welfare is equal to 150 because solar displaces 20 MW of gas, but the solar generator pays its fixed entry cost and the line upgrade costs 250. If the solar generator enters W, it produces 25 MW, earning $25 \times 30 - 200$. Welfare increases accordingly.

Immediately, this introduces a coordination problem: the solar generator only enters W conditional on the upgrade existing. But once the solar generator enters S, the upgrade reduces welfare relative to the no-upgrade world. Thus, achieving the first-best outcome requires jointly deciding on infrastructure provision and entry locations. This issue is present with wind generation too: solar generators faced a fairly gentle reduction in supply when locating in S. For wind generation in ERCOT, that penalty is much steeper, making entry unprofitable at any location without an upgrade.

Note that the results in this section are not unique to the electric grid. While a complex interaction of power flows leads to congestion on the electric grid, the same locational and technological distortion would flow through if electric transmission functioned like a pipeline, where flows could be directed by a system operator.

3 Entry Model

This section details the empirical model of generator entry I use. I treat generators as technology-specific, independently operated, single-plant firms.¹¹ I fix entrant sizes to the

¹¹Insert description of joint ownership in Texas, Johnston et al. (2024) find XXX.

average nameplate MW of wind and solar entrants from 2018-2024. Each period, generators make a single decision in the model: enter or permanently forgo the opportunity. Next, generators retire using an exogenous and reduced form process. I do not model generator exit because exit is rare.¹² In the baseline version of the model, I omit transmission line upgrades, reflecting the limited number of bulk transmission grid improvements ERCOT made over this period. I return to the upgrade process in counterfactuals. The final stage of the model is production, where the grid operator dispatches supply to meet demand. I describe the dispatch model in Section XX.

Entrants face site-specific entry opportunities, so the entry decision is at the region-technology level. Regions in the model are county clusters – there are 40 regions in total and each region has on average six counties within it.¹³ The number of generators of each technology τ eligible to enter each region r each period, $\bar{N}_{r\tau}$, is fixed at the largest number of region-year entrants seen in the region across all years plus one. Solar and wind generators made up the vast majority of entrants to the grid over the period I study, though battery storage entered in large quantities in 2023 and 2024. I treat new non-solar or wind entrants as exogenous shifts in supply.

Each period represents one year. The timing of the model proceeds as follows:

1. States and beliefs are realized. Entrants draw T1EV i.i.d. cost shocks and simultaneously decide on entry.
2. Retirement is realized. If chosen for retirement, generators exit. Generator entry occurs.
3. Production occurs for all studied hours using transmission capacity and supply curves constructed based on regional technology quantities. Payoffs accrue to generators and states transition.

Publicly observed state variables are given by $x_{r\tau}$:

$$x_{r\tau} = \{q_{r,\text{solar}}, q_{r,\text{wind}}, Q_{\text{solar}}, Q_{\text{wind}}\},$$

where r indicates the region in which an entrant is located, τ indicates the entrant's technology type, $q_{r,\text{solar}}$ indicates the quantity in MW of solar generation in r , $q_{r,\text{wind}}$ indicates

¹²Five wind resources exited during the 2018-2024 period, while zero solar resources did.

¹³Regions are generated based on edge characteristics, assessed on distance (smaller distances are better), capacity similarity (high-capacity corridors are grouped together), and congestion frequency (congestion separates regions). The goal is to cluster counties with limited intraregional congestion into a larger region. Implicitly, this says that when generators site in that region, intraregional deliverability is more robust than cross-region deliverability.

the quantity in MW of wind generation in r , and Q_τ indicates the total quantity in MW of each technology type on the grid. I omit the subscripts on x moving forward for clarity.

Instead of tracking the quantity of supply across all regions and for all technologies, entrants form beliefs over the allocation of Q_{solar} and Q_{wind} across the grid. g represents the true allocation of capacity. It determines LMPs, based on the spatial pricing mechanism that I discussed in Section 2. Entrants evaluate payoffs and transitions under the equilibrium conditional belief $M(g | x)$. The remaining beliefs depend on $p(x)$: q transitions according to realized entry counts and retirement. Q evolves under entry implied by $p(x)$.

The model is infinite-horizon and stationary. I assume that entrants play the same strategy each year conditional on the state. Demand in ERCOT rose by roughly 2% per year between 2018 and 2024. I solve for strategies using demand realizations from 2018 because sampling hours across years thus risks asking 2018-level capacity to meet 2024-level demand, which would lead to blackouts. There were also regional shifts in demand in ERCOT over this period. Oilfield electrification in the Far West led to localized demand doubling in the region and increased congestion. I cannot capture anticipatory entry in my model. I return to this in Section 5.

Changes to fixed costs (along with state variable transitions) drive new entry behavior each year. I treat these shifts as unexpected by entrants. The changes are driven by decreasing capital costs (solar) and shifting tax policy (wind). There is evidence that entrants may anticipate changes to tax policy, which could make it more difficult for me to pin down correct entry strategies (cite luming chen).

3.1 Entry

Each period t , a new set of entry opportunities $\bar{N}_{r\tau}$ appears in each region. Entrants decide whether they would like to pursue the entry opportunity $a_{r\tau} = 1$ or let it lapse $a_{r\tau} = 0$. Entry opportunities within a region-technology type are symmetric, thus all entrants play the same mixed strategy entry policy $p(x)$. Entrants must pay a fixed cost $F_{r\tau}$ to enter.

Entrants are forward-looking and make decisions about entry based on flow payoffs along with future grid evolution, implicitly trading off the resource quality of a location against its current and future expected transmission congestion. They observe the public moment state x and a vector of private shocks $\{\epsilon_{0r\tau}, \epsilon_{1r\tau}\}$, which I assume follow the T1EV distribution with scale parameter σ_τ . They do not condition their entry decision on the full other-region allocation g ; instead, they hold equilibrium beliefs $M(g | x)$ over the allocations consistent with the observed moment state. I discuss how entrants form $M(g | x)$ in Section 3.1.1.

Let β be the annual discount factor, common to entrants and incumbents. Let x' indicate

next period's state realization, $E_{p,M}^{inc}$ indicate the expectation for an incumbent over entry policy p and capacity allocation belief M and $E_{p,M}^{ent}$ indicate the expectation for an entrant. Let $\pi_{r\tau}$ denote the technology-specific flow payoff for a given x and g realization. Then, for a given entry policy p and allocation belief M , the value of a generator of type τ in region r on the grid $V_{r\tau}(x)$ and the value of entry $v_{r\tau,1}(x)$ are given by:

$$V_{r\tau}(x) = E_{p,M}^{inc}[\pi_{r\tau} + \beta V_{r\tau}(x')|x] \quad (1)$$

$$v_{r\tau,1}(x) = E_{p,M}^{ent}[\pi_{r\tau} + \beta V_{r\tau}(x')|x, a_{r\tau} = 1] \quad (2)$$

For a generator of technology τ in region r already on the grid, the value of being on the grid depends on the generator's production profile, which depends on τ , rival generator entry in all regions, given by p , and the generator's belief about how capacity is allocated across other regions, given by $M(g|x)$. Then, retirement is realized and dispatch is run, generating $\pi_{r\tau}$. The value of entering is almost identical, except the entrant needs to account for how its own entry affects state variables $q_{r\tau}$ and Q_τ and dispatch realizations.

Entrants weight the value of entry against fixed costs and cost shock realizations:

$$\tilde{v}_{r\tau,1}(x) = v_{r\tau,1}(x) - F_{r\tau} - \epsilon_{1r\tau} \quad (3)$$

$$\tilde{v}_{r\tau,0}(x) = -\epsilon_{0r\tau} \quad (4)$$

Let $p_{r\tau}(x)$ denote the entry probability for a technology- τ region- r opportunity in state x , and let $p \equiv \{p_{r\tau}\}_{\tau \in \mathcal{T}}$ denote the complete entry-policy profile. Projects enter when $\tilde{v}_{r\tau,1}(x) \geq \tilde{v}_{r\tau,0}(x)$. T1EV shocks yield the logit entry probability:

$$p_{r\tau}(x) = \frac{\exp([v_{r\tau,1}(x) - F_{r\tau}] / \sigma_\tau)}{1 + \exp([v_{r\tau,1}(x) - F_{r\tau}] / \sigma_\tau)}. \quad (5)$$

σ_τ controls the scale of the cost shock that entrants receive. Small σ_τ values indicate that entry is fairly deterministic conditional on the modeled value of entering in a region, while large σ_τ values indicate substantial unobserved heterogeneity in entry costs. I describe the process I use to estimate σ_τ for each technology in Section 3.1.2.

$p(x)$ gives the entry policy that generators play. Inverting Equation (5) yields a closed-form representation of fixed costs:

$$F_{r\tau} = v_{r\tau,1}(x) - \sigma_\tau \log\left(\frac{\hat{p}_{r\tau}(x)}{1 - \hat{p}_{r\tau}(x)}\right). \quad (6)$$

$\hat{p}(x)$ is observed entry frequencies. I then parameterize the model-implied fixed costs ac-

cording to observable fixed cost shifters, including capital costs, tax rebates, land costs, and a local Texas tax rebate that I discuss in Section 4. I aggregate capital costs, land costs, and tax rebates into a $C_{r\tau}^{\text{net capex}}$ value, while I include other non-price shifters in $Z'_{r\tau}$, which I discuss in more detail in Section 3.1.2. Letting θ_τ hold the coefficients for all non-price shifters, the parameterization is as follows:

$$F_{r\tau}(\theta_\tau) = C_{r\tau}^{\text{net capex}} + Z'_{r\tau}\theta_\tau. \quad (7)$$

The coefficients on this cost function are estimated using observed entry choices and model-generated entry values. The coefficient on net capex is fixed to one because it is measured in dollars. Projecting fixed costs onto observables provides a common fixed cost function for counterfactuals.

I rule out strategic delay in entry. There are several reasons this is appropriate in my setting. First, strategic delay requires the foregone revenue from not entering is smaller than any prospective declines in fixed costs. While fixed costs decreased over the period I study, revenues were also high. [insert yearly revenue, estimated fixed costs]...

I now describe the process by which beliefs are formed and the process by which I recover fixed costs in more detail.

3.1.1 Beliefs

Forecasting payoffs requires more than the observed states. Consider, for example, $q_{r,\text{solar}} \ll Q_{\text{solar}}$. If solar generation outside region r was uniform across all other regions, congestion would be low in region r . If outside solar generation was instead concentrated in a single region, congestion would be rampant on the grid, and prices in r would either be very low (if it was further from demand than the concentrated region) or very high (if it was closer to demand than the concentrated region). These situations, too, are not equally likely: other-region entry concentrating in a single region would be uneconomic. Entrants have beliefs over g , which is the full county-level allocation of generation. The belief over entry allocations $M(g|x)$ provides the distribution of g that is consistent with x for each x (Ifrach and Weintraub (2017)).¹⁴ In effect, entrants have a distribution of potential other region quantities for each Q that corresponds to how often that allocation of quantities would occur under the solved entry policy $p(x)$ and retirement.

In the equilibrium, M must coincide with the stationary allocation distribution of generation. Because p shifts entry probabilities, it shifts M : when equilibrium entry shifts in a given region, beliefs about other-region entry must also shift. To ensure this is the case,

¹⁴In equilibrium, M must coincide with the equilibrium entry allocation across regions.

beliefs are initialized at a random allocation. $\bar{N}_{r\tau}$ entrants then appear in each region. Each potential entrant receives a cost shock u and enters if $u < p(x)$. This follows the forward simulation method from Ifrach and Weintraub (2017).

Following entry decisions (and retirement), g transitions $g' = T(g_M, u; p)$; q transitions $q' = \sum_{c \in r} g'_c$; and Q transitions $Q' = \sum_r q'_r$. This yields x' . I update M , where the new M contains a dampened combination of original allocations and destination allocations observed following state transitions.¹⁵ Having a new M distribution shifts implied p and V values, thus restarting the values and entry policy loop. I continue estimating p , V , and M until a stationary equilibrium is reached.

3.1.2 Fixed Costs

The stationary equilibrium produces the region-technology-year threshold for fixed costs that aligns model and observed entry via equation (6). I parameterize the fixed cost threshold according to observable cost shifters using equation (7).

Using ordinary least squares to parameterize the fixed cost function is not well-behaved because many region-technology-year cells have no entry. Instead, I treat observed entry as data from a binomial process:

$$n_{r\tau t} \sim \text{Binomial}(\bar{N}_{r\tau}, p_{r\tau t}(\theta_\tau, \sigma_\tau)), \quad (8)$$

where θ_τ are the non-monetary parameters of the fixed cost function and σ_τ is the scale of the cost shock that entrants receive. Then, I select θ_τ such that model fixed costs maximize the binomial likelihood of the data:

$$\hat{\theta}_\tau(\sigma) = \underset{\theta_\tau}{\operatorname{argmax}} \sum_{r,t} [n_{r\tau t} \log p_{r\tau t} + (\bar{N}_{r\tau} - n_{r\tau t}) \log(1 - p_{r\tau t})]. \quad (9)$$

I estimate this likelihood separately for solar and wind entrants. For solar entrants, the specification includes net capex shifters and an intercept α_{solar} , with no other only non-cost shifters. For wind entrants, I augment the solar estimation equation with my measure of land costs.

Both solar and wind thus produce $\hat{F}_{r\tau}$, fixed costs that can be represented by shifters and fixed effects. Comparing the baseline model results to counterfactuals requires using consistent fixed cost parameterizations across each scenario; to facilitate this, I re-solve equilibrium using $\hat{F}_{r\tau}$.

¹⁵Note that $M(g|x)$ is conditional on x , so only allocations corresponding to the same state variables are included in the same distribution.

Equation (9) is a function of sigma. To estimate sigma, I test a grid of different sigma values ranging from $\sigma = 4$ to $\sigma = 4096$. For each technology and candidate σ , I proceed as follows.

I hold out one region r . Then, using all other regions, I estimate fixed cost coefficients:

$$\hat{\theta}_{\tau,-r}(\sigma) = \operatorname{argmax}_{\theta} \sum_{j \notin r, t} \ell(n_{j\tau t}, \bar{N}_{j\tau}; p_{j\tau t}(\theta, \sigma)). \quad (10)$$

I then use the fitted fixed cost function to predict entry in r and calculate the binomial log likelihood for the held out region. Then I repeat this until every region has been held out once and add together the likelihoods. I then select the σ that maximizes the sum of the likelihoods.

3.2 Retirement

I assign each generator an exit rate based on their technology type. Solar generators retire with probability 1/30 each year and wind generators retire with probability 1/25 each year. These hazards imply operating lives of 30 and 25 years, respectively, which correspond to the average lifespan of utility-scale wind and solar technology. I do not treat retirement as increasingly likely with generator age, primarily for computational reasons.¹⁶

This is primarily a technical assumption. Retirement is rare in sample – only five wind generators retired over the period I study – and arrives randomly. The shortest lived wind generator in ERCOT exited after 11 years of production, for example. Wind and solar marginal costs are zero in the model; modeling strategic retirement would be challenging. The retirement rates I use are sufficiently low that retirement is unlikely for any individual generator over the time span I look at. Additionally, adding retirement into the model prevents generation from accumulating indefinitely and makes the stationary distribution of generation less reliant on the model’s initial conditions.

3.3 Production

This subsection begins with a brief overview of economic dispatch on the electric grid, including key concepts about pricing and dispatch. It then proceeds by embedding these concepts within the production process for generators.

¹⁶An age-independent geometric lifetime is appealing because I do not have to keep track of generator ages.

3.3.1 Economic Dispatch

Let regions be denoted by r , dispatch periods denoted by t , and renewable technology denoted by τ . Region-technology-period production is given by $q_{r\tau t}$, total regional production by $q_{rt} \equiv \sum_{\tau} q_{r\tau t}$, total cost by $C_{r\tau t}(q_{r\tau t})$, and marginal cost by $MC_{r\tau t}(q_{r\tau t}) \equiv C'_{r\tau t}(q_{r\tau t})$. Regional demand is given by D_{rt} and system demand aggregates regional demand across regions $D_t = \sum_r D_{rt}$. Let flows on transmission lines be given by $\mathcal{F}_{ct}$, with transmission line capacities of K_c .

Grid operators solve a least cost dispatch optimization problem. For a given t , they choose generation of each technology (including non-renewable technologies) at each node on the grid to minimize costs. They face three constraints: system demand equals system generation, and flows on transmission lines do not exceed transmission line capacities. Each transmission line has a positive and negative flow constraint because flows are signed in the model. Satisfying the latter two constraints requires estimating how electricity flows across transmission lines. I discuss the framework for how this works in Section 2. In Appendix B, I discuss the process I use to construct a Power Transfer Distribution Factor (PTDF) Matrix for the full grid. The PTDF matrix calculates how electricity flows across the grid as injections and withdrawals of electricity from the grid change.

The grid operator's problem can be formulated as a Lagrangian:

$$\begin{aligned} \mathcal{L}_t = & \sum_r \sum_{\tau} C_{r\tau t}(q_{r\tau t}) + \lambda_t \left(\sum_r D_{rt} - \sum_r \sum_{\tau} q_{r\tau t} \right) \\ & + \sum_c \mu_t^{c,+} \left(\sum_r PTDF_{c,r}(q_{rt} - D_{rt}) - K_c \right) \\ & + \sum_c \mu_t^{c,-} \left(- \sum_r PTDF_{c,r}(q_{rt} - D_{rt}) - K_c \right). \end{aligned} \quad (11)$$

λ_t is a Lagrangian multiplier on the system demand and $\mu_t^{c,+}$ and $\mu_t^{c,-}$ are Lagrangian multipliers on transmission constraints. $\mu_t^{c,+}$ and $\mu_t^{c,-}$ are often referred to as the shadow price of transmission capacity, indicating the value of relaxing transmission constraints by one MW.¹⁷ Note that line constraints have both upper and lower limits, where sign depends on the direction in which power is flowing on the transmission line. For clarity, I omit generator production capacity limits.

Let the grid be uncongested, so $\mu_t^{c,+} = 0$ and $\mu_t^{c,-} = 0$ across all grid edges. Consider a marginal generator whose production is not at full capacity. Taking the derivative with

¹⁷Congestion costs, as discussed in Section 1, are calculated as $\sum_t \sum_c [\mu_t^{c,+} F_{ct} + \mu_t^{c,-} (-F_{ct})] \delta t$, summing the shadow price of the constraint over the period of time (δt) and in the direction in which it was constrained.

respect to $q_{r\tau t}$ yields the system price of electricity:

$$\frac{\partial \mathcal{L}_t}{\partial q_{r\tau t}} = MC_{r\tau t}(q_{r\tau t}) - \lambda_t = 0. \quad (12)$$

Let C_t^* be the minimized system dispatch cost and $D_t \equiv \sum_r D_{rt}$. Then the price of electricity in an uncongested system is set by the marginal cost of the marginal generator. By the envelope theorem, the cost of increasing demand by one MW is λ_t :

$$\lambda_t = \frac{\partial C_t^*}{\partial D_t}. \quad (13)$$

As the grid becomes congested, μ s turn on and local prices diverge from system prices. Taking the derivative with respect to $q_{r\tau t}$ yields the price of electricity in region r :

$$\frac{\partial \mathcal{L}_t}{\partial q_{r\tau t}} = MC_{r\tau t}(q_{r\tau t}) - \lambda_t + \sum_c (\mu_t^{c,+} - \mu_t^{c,-}) PTDF_{c,r} = 0. \quad (14)$$

Rearranging gives:

$$LMP_{rt} = \frac{\partial C_t^*}{\partial D_{rt}} = \lambda_t - \sum_c (\mu_t^{c,+} - \mu_t^{c,-}) PTDF_{c,r} = MC_{r\tau t}(q_{r\tau t}). \quad (15)$$

LMP_{rt} is called the locational marginal price for a given region.

3.3.2 Generator Production

Flow payoffs depend on expected revenues. Expected revenues are given based on the solution to the dispatch model. I describe how I estimate the dispatch model in Section B. Simplifying (15) yields the following representation of prices:

$$LMP_{rt} = \lambda_t + \text{congestion costs}_{rt}. \quad (16)$$

Regions on the demand-side of congestion have positive congestion costs, while regions on the generation-side of congestion have negative congestion costs.

I do not include all 8,760 hours of a year in the model. Instead, I estimate dispatch for 96 hours, which are chosen and weighted based on system representativeness.¹⁸ Let ω_t

¹⁸Hours differ in their level of demand, demand geography, time of year, time of day, and renewable supply. Renewable supply is based on empirical wind and solar availability data. Hours are weighted based on yearly representativeness – for example, though a high demand, low renewable supply period does not occur frequently, the revenues that generators see during these system conditions can account for a sizable portion of yearly revenues. Such an hour would be important to sample in the 96-hour sample, but would

denote the annual weight assigned to dispatch hour t , and let $q_{r\tau t}^*(x, g)$ denote the entrant's dispatched production. Its flow payoff is

$$\pi_{r\tau}(x, g) = \sum_t \omega_t q_{r\tau t}^*(x, g) LMP_{rt}(x, g), \quad (17)$$

Note that resource quality is embedded within $q_{r\tau t}^p$. If we let $q_{r\tau}$ indicate region-technology capacity and $f_{r\tau} \in (0, 1)$ indicate a region-technology pair's capacity factor, $q_{r\tau}^* \leq f_{r\tau} q_{r\tau}$.

3.4 Equilibrium

Equilibrium in this model is driven by the tradeoff between resource quality and dynamic congestion concerns. As congestion becomes more likely and more costly, this tradeoff becomes more acute.

Capturing these dynamics requires a representation of renewable capacity on the grid and where that capacity is located. With 40 regions, estimating a Markov Perfect Equilibrium for this setting would quickly become infeasible. Following Weintraub et al. (2008) and Ifrach and Weintraub (2017), I solve for a stationary moment-based Markov equilibrium (MME) where firms track the spatial allocation of rival capacity. Generators include details about their own region (renewable capacity) and a statewide summary of total renewable capacity according to their technology on the grid (Q_τ).

A stationary equilibrium in this model yields a triple (p^*, V^*, M^*) , where p^* gives entry probabilities in each region and for each technology and x ; V^* gives the value to a generator for being on the grid in each region and for each technology and x , and M^* gives beliefs about the allocation of generation across other regions for each x . I require that:

1. Given M^* and rival policy p^* , V^* satisfies the incumbent Bellman equation,
2. Entrant values constructed from V^* generate p^* based on the logit best response,
3. Entry, retirement, and stock transitions under p^* generate a stationary allocation distribution whose conditional law is $M^*(g | x)$.

Equilibrium computation requires the following steps:

1. Initialize M and p ;
2. Holding M fixed, solve the $V - p$ fixed point;
3. Simulate entry and retirement based on p ;

receive a low weight within that sampling.

4. Update $M(g \mid x)$ based on the simulated allocation of generation across regions;
5. Update V and p and repeat until policy, values, and the allocation law are stationary.

4 Data & Estimation

I compile three unique datasets to study entry empirically: an hourly production panel, which contains information on generators, including their supply curve and their location, and load, a measure of demand; a yearly infrastructure panel, which contains information on transmission lines, substations, and upgrades; and a yearly fixed cost panel, which contains information on capital costs, taxes, and other fixed cost shifters. Transmission capacity is unobserved; in Appendix B.1, I describe the process I use to recover transmission capacity and validation checks I perform to ensure my capacities are reasonable. I also describe the model I have for estimating commitment in Appendix B.1.¹⁹ I describe the data construction and validation for the capacity and commitment models further in Appendix XXX.

Production Panel Using Seasonal Assessment of Resource Adequacy (SARA) and Monthly Outlook for Resource Adequacy (MORA) reports, I compile a monthly panel of generators that includes information on generator county, technology, interconnection date, and nameplate capacity. I link this panel to ERCOT Security Constrained Economic Dispatch (SCED) data at the generator level. SCED data contains the supply curves that each generator submits to the grid operator every five minutes for constrained optimization, along with the amount of electricity they sold in a period. The supply curves reveal the amount of electricity each generator is willing to sell at a given price.

Load measures the volume of electricity that is demanded each hour. County-level load data for ERCOT is unavailable. I construct county-level load based on ERCOT’s hourly zonal load data (NP6-345-CD). I describe the process I use to construct county-level load in Appendix XX.

Infrastructure Panel The infrastructure panel is based on transmission line data, which comes from the Homeland Infrastructure Foundation Level Data (HIFLD). It is a database of transmission lines, ranging in voltage from 69 kV to 765 kV, available in the U.S. It contains information on the geographic location of transmission lines and the voltage of those lines. Transmission line capacity is proportional to voltage. To focus on bulk transmission of electricity, I restrict attention to ERCOT’s 230 and 345 kV transmission lines, which are

¹⁹Each hour, generators (and sometimes the grid operator) commits to produce during that hour. Committed generators produce electricity under all load realizations, which reduces the amount of load that renewable generators must meet and changes prices.

ERCOT’s highest capacity lines and most frequently used for bulk transmission.²⁰ I link transmission line data to line-level congestion information available from ERCOT (NP6-86-CD). ERCOT’s congestion data reports, at the 5-minute level and between substation pairs, where congestion arose on the grid. It also reveals the transmission capacity in MW between the substations where the congestion happened and the shadow price of that congestion, which I describe in Section B.

I link ERCOT’s congestion data to physical grid segments via documentary review, which I describe in Appendix XX. I use the linked dataset to identify where congestion arose at the county pair level. For the remainder of the analysis, I condense the bulk transmission kV grid in ERCOT into a graph, where counties are nodes and any transmission line that runs through two counties is treated as an edge. I assume that electricity can be stepped down from the bulk transmission grid in any county that has a transmission line running through it. In many situations, multiple transmission lines connect two counties. I recover the implied joint capacity for these transmission lines. Implicitly, I treat intracounty electricity transmission as freely available. A limited number of counties are islanded, meaning they do not have any 230 or 345 kV transmission lines. I assume generation from those counties is freely transmitted to the bulk transmission backbone in proportion to the total number of 138 kV lines that connect the islanded county to counties that lie on the transmission backbone. In total, there are 204 county nodes connected by 300 edges. 10 nodes do not have any 230 or 345 kV transmission lines.

Fixed Cost Panel I construct a panel of yearly fixed cost shifters for both wind and solar generators. I treat capital costs, land costs, and state and federal taxes as fixed cost shifters, which are all either time- or region-varying. I cannot measure some things that clearly contribute to fixed costs, like interconnection fees or labor costs. This omitted variation is not separately identified and loads into the regional fixed-cost intercept that I estimate, described in Section 3.

I use capital costs from the Lawrence Berkeley National Laboratory 2025 wind and solar market reports (Utility-Scale Solar Update and Land-Based Wind Energy Technology Update). I calculate the price of land based on the Texas A&M Real Estate Research Center’s annual large-tract land price database. I measure the Texas Chapter 313 policy tax abatement that generators were eligible to enter into with local school districts, a tax incentive for renewable generators.²¹ I apply the Production Tax Credit (PTC) to wind generators and

²⁰In two instances, I found that ERCOT described transmission lines as 345 kV lines while the HIFLD characterized them as 138 kV lines. I assumed that ERCOT’s characterization was correct in this case.

²¹The Chapter 313 abatement was a property tax incentive for capital investments, where investor property taxes were capped for school district Maintenance and Operations taxes purposes. Maintenance and Operations taxes vary at the school district level. The program expired at the end of 2022, though abate-

the Investment Tax Credit (ITC) to solar generators. The PTC applies to wind production; I annualize PTC credits to generators based on county-level wind capacity factors and sum PTC credits over a 10-year horizon.²² PTC credit rates varied over the 2018-2025 period: in 2018, the rate was \$0.018/kWh; in 2019, \$0.012/kWh; in 2020-2021, \$0.018/kWh; and in 2022-2024, \$0.03/kWh. The ITC is calculated as the ITC rate times the capital cost of solar projects in a given year. The ITC rate over this period was 30% in all years except 2020 and 2021, when the ITC rate was 26%.

4.1 Validation

Accurately capturing entry incentives on the grid requires an accurate representation of prices that entrants would see under counterfactual entry. I performed a validation exercise to assess whether the dispatch model (along with the capacity and commitment models) produce realistic prices. The exercise involved estimating the dispatch model on 2,000 hours from 2023, which were excluded from the fit of the model.

Table 3: Out-of-Sample Validation of the Dispatch Model

	Model	Observed	Bias	MAE	RMSE	Correlation
<i>Panel A: System price validation</i>						
Demand-weighted LMP (\$/MWh)	52.38	52.24	0.14	29.14	192.30	0.85
Load-shedding frequency	0.00	—	—	—	—	—
<i>Panel B: Renewable price-value validation</i>						
Solar capture price (\$/MWh)	71.89	67.67	4.22	—	—	—
Wind capture price (\$/MWh)	25.26	22.98	2.28	—	—	—
Solar revenue (\$/MW-year)	116,078	109,775	6,303	13,715	22,089	0.86
Wind revenue (\$/MW-year)	64,378	58,567	5,811	13,674	19,285	0.61

Notes: The table evaluates the dispatch model on 2,000 evenly sampled hours from 2023, which are excluded from estimation of the model. Revenue statistics are annualized by weighting each sampled hour by 8,760/2,000. The solar and wind statistics average county-level per-MW outcomes across 23 and 44 electrical county nodes, respectively. Bias is the average difference between Model and Observed outcomes. MAE is the mean average error. RMSE is sensitive to a small number of scarcity-price observations.

The model shows very strong fit at the system price and renewable price levels. Model demand-weighted LMPs in 2023 are \$52.38, while observed demand-weighted LMPs are \$52.24. The Mean Absolute Error between model and observed prices is higher – \$29.14 – though much of that discrepancy is due to tail events driving up mean. Median absolute

ments negotiated before 2022 were allowed to continue until their original expiry date occurred. I generate an annual value of this program for each project with a negotiated agreement.

²²The next time I estimate the model I will shift this to a net present valuation.

error, for example, is \$3.89. The correlation across prices is also very high – 85%. Note that RMSE is high because of tail events, which are difficult to price accurately. Tail events are important for accurately estimating entrant revenues, as they have the potential to increase revenue dramatically. The difficulties pinning down tail revenues do not cause issues for expected wind and solar yearly per-MW revenues, which are quite accurate.

4.2 Estimation

This section describes estimation of the dynamic entry model. The model seeks to construct entry values such that they rationalize observed entry choices at the region-technology-year level. The estimation process is summarized in Figure 3. The relevant objects in this section are as follows: p , entry probabilities; V , an entrant’s (or incumbent’s) value of being on the grid; $M(g|x)$, an entrant’s beliefs over the allocation of renewable capacity across other regions; F , the fixed costs of entry; and σ , the scale of entrant cost shocks. I solve two nested fixed points in order to estimate the model. In the first fixed point, I solve for equilibrium based on flow profits from the dispatch model and entry probabilities from finding a fixed point in V - p . In the second fixed point, I add in entrant beliefs over $M(g|x)$, which must be accurate in equilibrium. Then, I recover fixed cost thresholds. I choose σ based on fixed cost model fit over a grid of σ values. Then I project fixed costs onto the fixed cost panel shifters to obtain a representation of fixed costs that I can take to counterfactuals. Doing so requires me to re-estimate the nested fixed point routine.

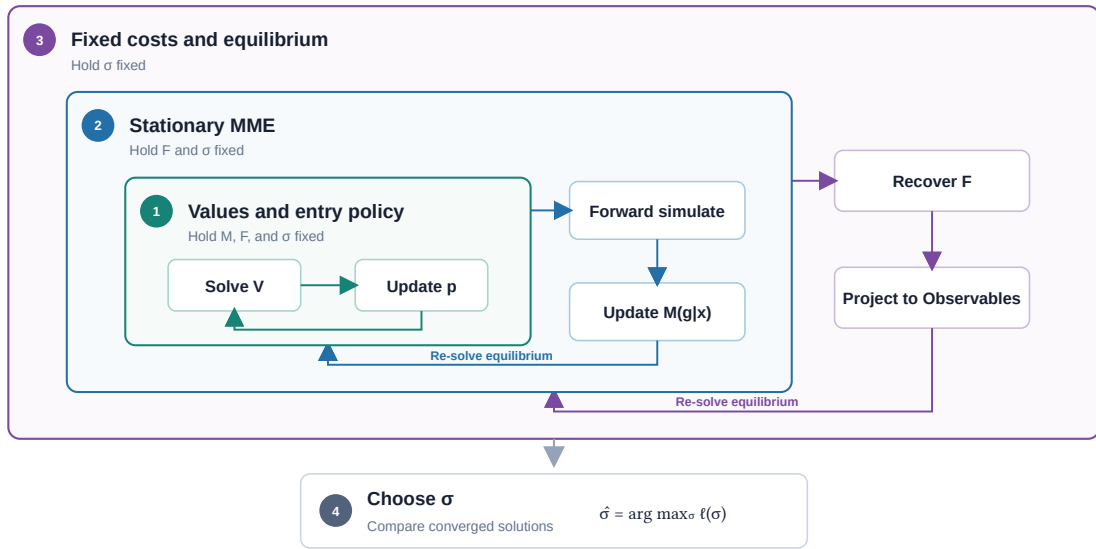


Figure 3: Estimation Flow

5 Results

This section details the results from the dynamic entry model. I describe the estimated fixed costs $\bar{F}_{r\tau}$ and the model-implied project values $V_{r\tau}$. I then look at the spatial allocation of entry for the model relative to the observed spatial allocation of entry.

5.1 Model Values

Estimated project costs $\bar{F}_{r\tau}$ and model-implied project values $V_{r\tau}$ are shown in Table 4. The table aggregates values across regions and across time periods. Costs and values are reported in MUSD/MW. I find that fixed costs for wind generators are almost three times larger than they are for solar generators. The distribution of fixed costs for solar is almost uniform across regions, whereas wind fixed costs vary significantly across regions. This is because of the fixed cost specification, where wind fixed costs are discounted in regions with pre-existing wind entry. Conceptually, this is consistent with a model where there are significant regional benefits to pre-existing wind, like developers already working in the area, streamlined permitting processes, and lesser interconnection fees. I assume that those regional benefits do not extend to solar generators. For both wind and solar, fixed costs fall during the period I study. For solar, this is driven by sharp capital cost decreases, while for wind it is primarily driven by tax benefits.

Fixed cost results imply average total fixed costs of \$115 MUSD for solar generators and \$443 MUSD for wind generators. A 2023 EIA report estimated installed construction costs for solar at \$196 MUSD and wind at \$294 MUSD respectively (U.S. Energy Information Administration (2025)).²³ The fixed cost parameterization anchors both technologies to net capital costs. Wind fixed costs additionally vary with the regional land-cost proxy, while solar fixed costs include only the net-capital-cost anchor and an intercept.

Table 4 also reports implied $V_{r\tau}$ for wind and solar. I find that mean $V_{r\tau}$ is lower than fixed costs for wind and solar, consistent with entry rates of 5% and 13% respectively. Mean $V_{r\tau}$ is roughly two-thirds the size of average fixed costs for solar and one-third the size of fixed costs for wind. Project values for wind are higher than they are for solar. Intuitively, $V_{r\tau}$ depends on (and increases in) the amount of electricity that a generator produces over its lifetime and the price it receives for that electricity. Wind generators have higher capacity factors (a measure how much electricity a generator can produce relative to its capacity) than solar generators do, which drives the observed differences in $V_{r\tau}$. This is true even though wind generators face lower prices than solar generators do: on average, LMPs for wind generators are 4.3% below the system price of electricity, whereas they are only 0.9%

²³Seel et al. (2024) and Wiser et al. (2024) estimate similar fixed costs for wind and solar.

Table 4: Estimated Project Costs and Dynamic Project Values

	Solar	Wind
<i>$\bar{F}_{r\tau}$, estimated project costs</i>		
Mean $\bar{F}_{r\tau}$	0.75	2.10
p10 $\bar{F}_{r\tau}$	0.59	1.06
p90 $\bar{F}_{r\tau}$	1.10	3.28
<i>$V_{r\tau}$, model-implied project values</i>		
Mean $V_{r\tau}$	0.47	0.64
Region-year SD of $V_{r\tau}$	0.06	0.13
p10 $V_{r\tau}$	0.38	0.47
p90 $V_{r\tau}$	0.54	0.82
Mean congestion discount (percent)	0.9	4.3
Region-year SD of congestion discount (p.p.)	2.1	6.1

Notes: All monetary entries are reported in MUSD/MW and are evaluated over observed 2018–2024 regional states. Statistics weight region-year cells equally. The congestion discount is the percent reduction in project value under local rather than system prices.

below the system price for solar generators.²⁴ The discount faced by wind generators varies dramatically; in one region (Willacy county, in the far south of Texas), average congestion discounts faced by entrants are 24.7%.

Figure 4 assesses the spatial fit of the dynamic entry model. Each point in the figure represents a model region. The x-axis shows observed cumulative entry in representative-project equivalents for each region between 2018 and 2024, and the y-axis shows corresponding model-implied cumulative entry.²⁵ The correlation between model and observed entry is 0.81 for solar and 0.91 for wind. A line of best fit has slope 0.34 for solar and 0.64 for wind. The model compresses the spatial distribution towards the mean for both wind and solar.

There are several points where observed wind and solar entry is higher than model-predicted wind and solar entry. The far west is difficult for both wind and solar to pin down correctly: in the data, 18 wind generators and 23 solar generators entered the far west, while the model predicts only 13.5 and 11.5 generators enter in that region. One potential reason for this comes from Table 5: the far west is where demand grew most dramatically over this period, and thus where one might expect the stationary demand assumption to be more problematic. Other regional misses are less symptomatic: wind generator entry is too low in the eastern plains of Texas; solar entry is too low in west-central Texas.

To understand the magnitude of the congestion and resource quality channels, I estimate

²⁴All results in Table 4 are conditional on entry.

²⁵Including an intercept in the fixed effect parameterization mechanically recovers entry in the aggregate.

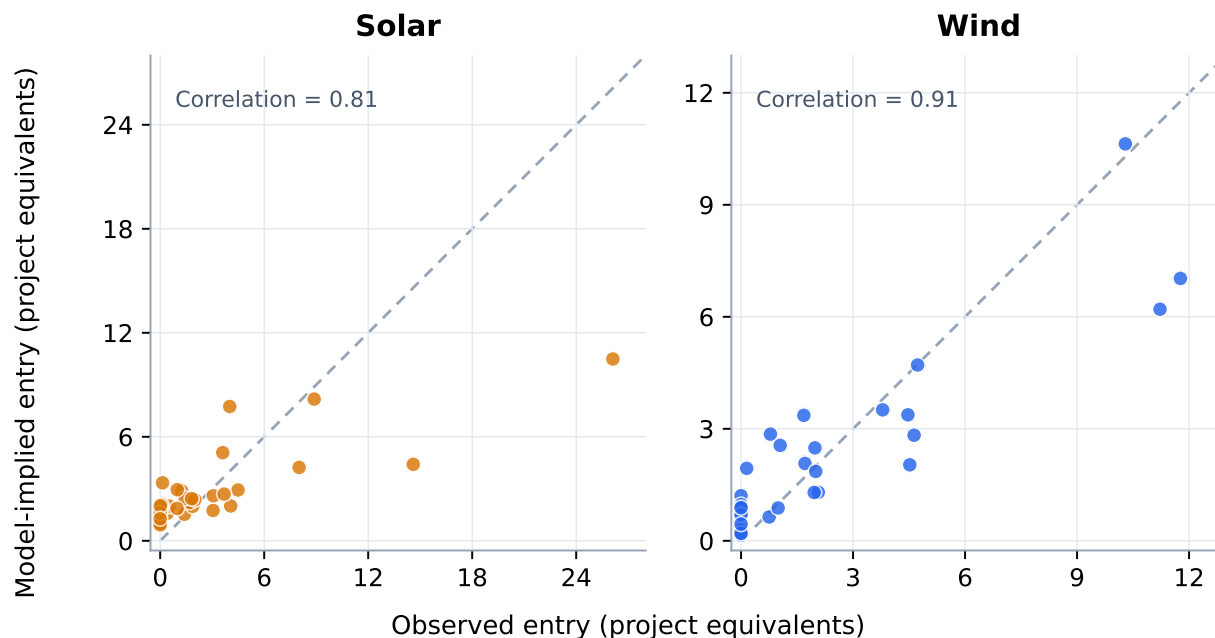


Figure 4: Observed and Model-Implied Cumulative Regional Entry, 2018–2024

Notes: Each point is a model region. The dashed line is the 45-degree line.

entry under two alternative scenarios. In the first scenario, I set congestion costs equal to zero everywhere on the grid and allow entry to adjust. Note that I do not estimate a new equilibrium. Figure 5 reports both the change in total entry and the change in its regional distribution. I find that wind entry increases by 8.9%, a total of 8.2 project equivalents. The new distribution of entry is mostly centered on the original distribution of entry: the distance between baseline model entry shares and no-congestion entry shares is 1.9%. Solar entry changes are more modest: total entry increases by 1.8% (2.2 project equivalents) and the distance between shares is 1.1%. Thus, eliminating congestion primarily affects the level of wind entry, with comparatively small changes in the spatial allocation of either technology.

Next, I examine how entry changes if resource quality were to shift. First, I give wind generators solar generators' resource quality distribution and geography, rescaled to preserve wind's maximum resource quality. I find this increases wind entry by 2.9% (2.7 project equivalents), while shares differ by 2.8%, indicating that adopting solar's resource geography substantively changes where wind generators locate. In the opposite exercise, adopting wind's resource geography reduces solar entry by 3.8% (4.6 project equivalents) while shares differ by 7.0%. Thus, wind's resource geography substantially changes where solar enters even though the decline in total solar entry is relatively modest.

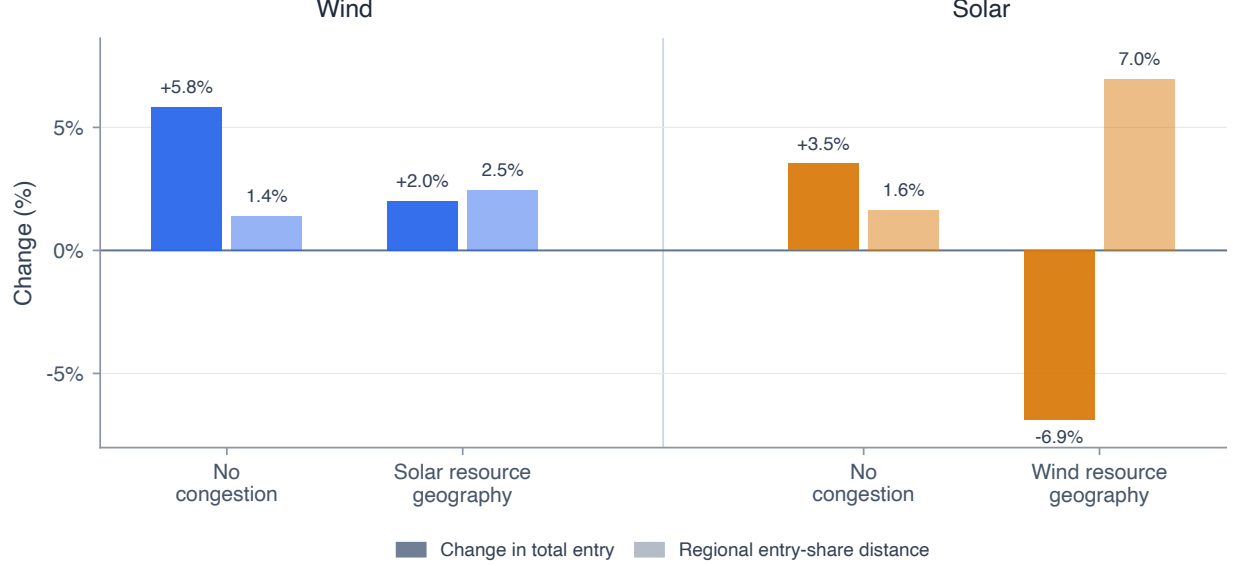


Figure 5: Entry Levels and Spatial Distributions under Alternative Scenarios

Notes: Dark bars report partial-equilibrium changes in total entry relative to the estimated baseline. Light bars report the total-variation distance between the baseline and counterfactual regional entry-share distributions: one-half the sum of the absolute regional share changes.

6 Counterfactuals

The entry model illustrates that congestion shifts generator entry decisions. It remains unclear how costly the lack of coordination between generation and transmission is. In a first counterfactual exercise, I test two different infrastructure provision policies, each designed to target the coordination inefficiency. To assess the effect of new policies, I initialize the model in 2018, implement the new policy, and compute an equilibrium of the model under the new policy until 2024. My outcome of interest is social welfare, which I calculate as:

$$W_k = -\mathbb{E}_k \left[\sum_{h \in \mathcal{H}} \omega_h \left\{ \sum_r \sum_{\tau} C_{r\tau h}(q_{r\tau h k}) + \chi e_{hk} \right\} + a_G \sum_{i \in \mathcal{G}_k} F_i + a_T K(\Delta Z_k) \right]. \quad (18)$$

Here, h indexes hour of the year, k indexes the policy, e_{hk} denotes carbon dioxide emissions, and χ is the social cost of carbon. $\mathcal{G}_k$ is the set of projects that enter under policy k , and F_i is project i 's fixed cost from the model. $K(\Delta Z_k)$ is the cost of transmission construction. a_G and a_T are the annualization factors for generation and transmission investment, respectively. Roughly, welfare depends on three objects: production costs (generation, emissions); new entrant fixed costs; and transmission costs. Note that demand is not included in

welfare because it remains fixed across policies.

The first policy, which I call the upgrade-on-entry policy, adapts an entry policy used in much of the rest of the U.S. to the Texas grid.²⁶ In it, grid operators add an intermediate stage between generators entering and production. In this stage, grid operators run the dispatch model over three different planning hours for the entrants. The hours are chosen to mimic high system-stress conditions – I use data from an hour with high load, one wind supply is particularly high relative to load, and one where solar supply is particularly high relative to load. If the entering generator causes congestion in those hours, then they are required to pay entry fees to provide the least-cost upgrade to the grid that prevents congestion from arising. The entry fees are split in proportion to the amount of the congestion that they cause on each line. This policy forces generators to internalize the cost that their entry imposes on the grid and allows them to choose locations with minimal congestion risk.

Next, I consider a market for new transmission. This policy is similar to “open season” policies used for new pipeline development; (Kavulla, 2026) calls for implementing a similar idea in response to data center demand. Under this policy, grid operators add an intermediate stage between generations entering and production. In this stage, entrants, incumbents, and load all submit bid curves to the grid operator, revealing their willingness-to-pay for new transmission. The grid operator then selects the upgrade (or set of upgrades) that maximizes willingness-to-pay minus upgrade costs.²⁷ This policy targets the coordination inefficiency directly: agents internalize both the benefits and costs of new transmission in proportion to their use of that transmission.

6.1 Model Additions

The counterfactuals require two additions to the model: a method for selecting upgrades and state variables that contain information about grid capacity. I discuss each of them in turn.

There are 300 corridors in the dispatch model, each of which is eligible for an upgrade. I treat upgrades as a flat 1200MW capacity addition to a corridor. There are significant complementarities across upgrades. For example, if a rural, high-wind region is connected to demand by a single transmission line that has two corridors, upgrading one of those corridors does not improve the grid. Congestion still arises on the unupgraded corridor. Instead,

²⁶The policy is often called “invest-and-connect,” whereas the Texas policy is called “connect-and-manage.” Invest-and-connect policies famously lead to long wait times in interconnection queues, see Johnston et al. (2024), an issue I do not consider in this paper.

²⁷Bids are constrained to be positive. I assume bids are honest and accurate; designing an incentive-compatible auction for new transmission is a future direction of research.

properly evaluating the value of an upgrade requires evaluating the effect of upgrading both links at the same time.

To do so, I proceed with a beam search through upgrade packages (add cite). Beam search, which comes from computer science, is a graph and tree search algorithm that focuses on searching near high-value nodes. For example, consider a network with four corridors: A, B, C , and D . A beam search begins by evaluating an upgrade on each individual corridor. It then identifies the W top upgrades and moves on to the next level of evaluation. Consider for example, $W = 2$, where A and C survive. Then the search evaluates upgrade pairings AB, AC, AD, BC, CD , but not BD . The search continues moving through levels until potential upgrade groups are exhausted.²⁸

For the purposes of the counterfactual, only corridors with congestion or corridors which reduce congestion elsewhere following an upgrade are included in the beam search. Any corridor upgrade that only becomes valuable conditional on another upgrade occurring and that also is not originally congested will not be included in the search. Assessing a corridor in the beam search involves applying a 1200MW capacity addition to the corridor, re-computing the PTDF matrix, and seeing how electricity flows adjust to determine whether the upgrade changes congestion on the grid. The width W of the search at each level is 100.

The upgrades chosen in equilibrium shift congestion and therefore prices. Including the grid as a state would be computationally infeasible. Instead, I model generators as caring about upgrades insofar as they change local prices, curtailment, and profits. This effectively treats the grid as a hidden state, similar to the way that entrants take other-region entry into consideration. As a result, counterfactual beliefs $M(g, Z|x)$ are formed over the spatial allocation of entry g and the capacity of the grid Z . I solve for beliefs using forward simulation, as in Section 3.1.1.

Because generators only consider the effect of upgrades on prices, curtailment, and profits, they do not submit bids for individual upgrades. Instead, they submit bids for deliverability, which is their average curtailment rate across all hours in the entry model. Note that, because entrants have beliefs over upgrades and entry, bid curves are not monotonically increasing: sometimes, increasing deliverability means increasing market integration such that prices decline, and entrant willingness-to-pay declines accordingly.

6.2 Results

TK

²⁸Note that BD is never evaluated in this example; exhausting potential upgrade pairings mean all pairings which include either A or C . All three- and four-level groups including A and C would be evaluated.

6.3 Data Center Experiment

TK

7 Conclusion

References

- American Electric Power (2008). “Transmission Facts.” URL https://web.ecs.baylor.edu/faculty/grady/_13_EE392J_2_Spring11_AEP_Transmission_Facts.pdf.
- Arkolakis, Costas and Conor Walsh (2023). “Clean Growth.” *Working Paper*, URL <https://static1.squarespace.com/static/5b4b997eda02bc592b5131cd/t/64e38ceeb5dae75b3b9cb7e3/1692634362860/CleanGrowth.pdf>.
- Boatwright, Abigail (2025). “Transmission Congestion is Knocking the Wind Out of Renewable Investments.” *Working Paper*.
- Borenstein, Severin, James Bushnell and Steven Stoft (2000). “The Competitive Effects of Transmission Capacity in a Deregulated Electricity Industry.” *RAND Journal of Economics*, 31(2): 294–325, URL <https://www.jstor.org/stable/2601042>.
- Brancaccio, Giulia, Myrto Kalouptsi and Theodore Papageorgiou (2024). “Investment in Infrastructure and Trade: The Case of Ports.” Working Paper 32503, National Bureau of Economic Research, URL <https://doi.org/10.3386/w32503>.
- Canni re, Charlotte De (2026). “Market Integration and Market Efficiency: Evidence from Transmission Constraints in the Belgian Electricity Sector.” *Working Paper*, URL <https://drive.google.com/file/d/11ThfDoWiZRIfd-G1dhqwULZrV0C6BoUC/view>.
- Cicala, Steve (2022). “Imperfect Markets versus Imperfect Regulation in US Electricity Generation.” *American Economic Review*, 112(2): 409–441, URL https://www.stevecicala.com/papers/elec_gov_v_mkt/elec_gov_v_mkt_draft_2.pdf.
- Cohn, Julie A. (2022). “Connecting Past and Future: A History of Texas’ Isolated Power Grid.” Research Paper 12.01.22, Center for Energy Studies, Rice University’s Baker Institute for Public Policy, URL <https://www.bakerinstitute.org/research/connecting-past-and-future-history-texas-isolated-power-grid>.
- Congressional Budget Office (2026). “Overview of Electric Transmission.” URL <https://www.cbo.gov/publication/62705>, presentation published August 27, 2026.
- Dunlop, R. D., R. Gutman and P. P. Marchenko (1979). “Analytical Development of Loadability Characteristics for EHV and UHV Transmission Lines.” *IEEE Transactions on Power Apparatus and Systems*, PAS-98(2).

- Duranton, Gilles and Matthew A. Turner (2011). “The Fundamental Law of Road Congestion: Evidence from US Cities.” *American Economic Review*, 101(6): 2616–2652, URL <https://doi.org/10.1257/aer.101.6.2616>.
- Electric Reliability Council of Texas (2025). “Fuel Mix Report: 2007–2024.” URL https://www.ercot.com/files/docs/2021/03/10/FuelMixReport_PreviousYears.zip, fuel mix reports for 2007–2024.
- Electric Reliability Council of Texas (2026). “About ERCOT.” URL <https://www.ercot.com/about>, accessed August 27, 2026.
- Fell, Harrison, Daniel T. Kaffine and Kevin Novan (2021). “Emissions, Transmission, and the Environmental Value of Renewable Energy.” *American Economic Journal: Economic Policy*, 13(2): 241–272, URL <https://www.aeaweb.org/articles?id=10.1257/pol.20190258>.
- Fernald, John G. (1999). “Roads to Prosperity? Assessing the Link between Public Capital and Productivity.” *American Economic Review*, 89(3): 619–638, URL <https://doi.org/10.1257/aer.89.3.619>.
- Golden, Dana (2026). “Investment and the Transfer of Power: Dynamic Effects of Transmission in Electricity Markets.” *Working Paper*, URL [https://jgolden36.github.io/files/JMP%20\(11\).pdf](https://jgolden36.github.io/files/JMP%20(11).pdf).
- Gonzales, Luis E., Koichiro Ito and Mar Reguant (2023). “The Investment Effects of Market Integration: Evidence from Renewable Energy Expansion in Chile.” *Econometrica*, 91(5): 1659–1693, URL https://koichiroito.com/pdfs/Market_Integration_Gonzales_Ito_Reguant.pdf.
- Gowrisankaran, Gautam, Stanley S. Reynolds and Mario Samano (2016). “Intermittency and the Value of Renewable Energy.” *Journal of Political Economy*, 124(4): 1187–1234, URL https://www.nber.org/system/files/working_papers/w17086/revisions/w17086.rev2.pdf.
- Hausman, Catherine (2024). “Power Flows: Transmission Lines, Allocative Efficiency, and Corporate Profits.” *American Economic Review*, URL <https://www.nber.org/papers/w32091>.
- Hogan, William, Juan Rosellón and Ingo Vogelsang (2010). “Toward a Combined Merchant-Regulatory Mechanism for Electricity Transmission Expansion.” *Journal of Regula-*

- tory Economics*, 38: 113–143, URL <https://link.springer.com/article/10.1007/s11149-010-9123-2>.
- Holland, Stephen P., Erin T. Mansur and Andrew J. Yates (2024). “Electrification in the Long Run.” *Working Paper*, URL https://mansur.host.dartmouth.edu/papers/holland_mansur_yates_longrun.pdf.
- Ifrach, Bar and Gabriel Y. Weintraub (2017). “A Framework for Dynamic Oligopoly in Concentrated Industries.” *Review of Economic Studies*, 84(3): 1106–1150, URL <https://doi.org/10.1093/restud/rdw047>.
- Jankovska, Olivera and Julie A. Cohn (2020). “Texas CREZ Lines: How Stakeholders Shape Major Energy Infrastructure Projects.” Research paper, Rice University’s Baker Institute for Public Policy, URL <https://www.bakerinstitute.org/research/texas-crez-lines-how-stakeholders-shape-major-energy-infrastructure-projects>.
- Jia, Panle (2008). “What Happens When Wal-Mart Comes to Town: An Empirical Analysis of the Discount Retailing Industry.” *Econometrica*, 76(6): 1263–1316, URL <https://onlinelibrary.wiley.com/doi/abs/10.3982/ECTA6649>.
- Johnston, Sarah, Yifei Liu and Chenyu Yang (2024). “An Empirical Analysis of the Interconnection Queue.” *Working Paper*, URL https://econ-chenyu-yang.github.io/interconnection_queues.pdf.
- Joskow, Paul and Jean Tirole (2005). “Merchant Transmission Investment.” *Journal of Industrial Economics*, 53(2): 233–264, URL <https://onlinelibrary.wiley.com/doi/abs/10.1111/j.0022-1821.2005.00253.x>.
- Kavulla, Travis (2026). “How Will Data Centers Pay for Power?” *American Affairs*, X(2), URL <https://americanaffairsjournal.org/2026/05/how-will-data-centers-pay-for-power/#notes>.
- Léautier, Thomas-Olivier (2000). “Regulation of an Electric Power Transmission Company.” *The Energy Journal*, 21(4): 61–92, URL <https://www.jstor.org/stable/41322900>.
- Mays, Jacob (2023). “Generator Interconnection, Network Expansion, and Energy Transition.” *IEEE Transactions on Energy Markets, Policy and Regulation*, 1(4): 410–419, URL <https://ieeexplore.ieee.org/document/10121623>.
- North American Electric Reliability Corporation (). “FAC-008-5: Facility Ratings.” URL <https://www.nerc.com/pa/Stand/Reliability%20Standards/FAC-008-5.pdf>.

- Seel, Joachim, Julie Mulvaney Kemp, Anna Cheyette, Dev Millstein, Will Gorman, Seongeun Jeong, Dana Robson, Rachman Setiawan and Mark Bolinger (2024). “Utility-Scale Solar, 2024 Edition: Empirical Trends in Deployment, Technology, Cost, Performance, PPA Pricing, and Value in the United States.” Technical report, Lawrence Berkeley National Laboratory, URL <https://emp.lbl.gov/publications/utility-scale-solar-2024-edition>.
- Seim, Katja (2006). “An Empirical Model of Firm Entry with Endogenous Product-Type Choices.” *RAND Journal of Economics*, 37(3): 619–640, URL <https://pages.stern.nyu.edu/~acollard/seim.pdf>.
- Shreve, Nathan, Julia Selker, Zach Zimmerman and Richard Doying (2025). “Transmission Congestion for 2024.” Report prepared for the connected grid initiative, Grid Strategies LLC, URL https://gridstrategiesllc.com/wp-content/uploads/GS_Transmission-Congestion-for-2024.pdf.
- Sweeting, Andrew (2013). “Dynamic Product Positioning in Differentiated Product Markets: The Effect of Fees for Musical Performance Rights on the Commercial Radio Industry.” *Econometrica*, 81(5): 1763–1803, URL <https://doi.org/10.3982/ECTA9856>.
- Trabish, Herman K. (2025). “Will ERCOT’s Streamlined Connect-and-Manage Approach Work for Other Markets?” *Utility Dive*, URL <https://www.utilitydive.com/news/ercot-connect-and-manage-spp-miso-eris/749083/>.
- U.S. Energy Information Administration (2025). “Construction Cost Data for Electric Generators Installed in 2023.” URL <https://www.eia.gov/electricity/generatorcosts/archive/2023/>, data collected on Form EIA-860, Annual Electric Generator Report.
- U.S. Energy Information Administration (2026). “Form EIA-860 Detailed Data with Previous Form Data (EIA-860A/860B).” URL <https://www.eia.gov/electricity/data/eia860/>, form EIA-860 detailed data, including historical EIA-860A and EIA-860B files and the 2025 early release.
- Vogelsang, Ingo (2001). “Price Regulation for Independent Transmission Companies.” *Journal of Regulatory Economics*, 20: 141–165, URL <https://link.springer.com/article/10.1023/A:1011103328722>.
- Weintraub, Gabriel Y., C. Lanier Benkard and Benjamin Van Roy (2008). “Markov Perfect Industry Dynamics With Many Firms.” *Econometrica*, 76(6): 1375–1411, URL <https://doi.org/10.3982/ECTA6158>.

- Wiser, Ryan H., Dev Millstein, Ben Hoen, Mark Bolinger, Will Gorman, Joseph Rand, Galen L. Barbose, Anna Cheyette, Naïm R. Darghouth, Seongeun Jeong, Julie Mulvaney Kemp, Eric O’Shaughnessy, Bentham Paulos and Joachim Seel (2024). “Land-Based Wind Market Report: 2024 Edition.” Technical report, Lawrence Berkeley National Laboratory, URL <https://emp.lbl.gov/publications/land-based-wind-market-report-2024>.
- Wolak, Frank A. (2015). “Measuring the competitiveness benefits of a transmission investment policy: The case of the Alberta electricity market.” *Energy Policy*: 426–444, URL https://fawolak.org/pdf/alberta_transmission_wolak.pdf.
- Wollmann, Thomas G. (2018). “Trucks without Bailouts: Equilibrium Product Characteristics for Commercial Vehicles.” *American Economic Review*, 108(6): 1364–1406, URL <https://doi.org/10.1257/aer.20160863>.

A Data Details

Table 5 shows summary statistics for the production panel. Average prices increase over the 2018-2025 period; very high prices in 2021 are due to Winter Storm Uri, which hit ERCOT in February and caused widespread chaos throughout the grid, with several important transmission facilities failing, renewables unable to deliver electricity to consumers, and gas lines freezing. Prices were as high as \$10,000/MWh during the storm. Generation and demand increased almost 25% over the period, with most of the additions coming in the 2022-2024 range. Curtailment increased over the period as well, as more renewables competed for a roughly fixed amount of transmission capacity. Regionally, the Far West had the most growth, with demand more than doubling between 2018 and 2024. Demand is otherwise concentrated in load centers like the Coast, North Central, and South Central. Generation is more evenly spread out, and renewable curtailment is most frequent in regions where generation outstrips demand.²⁹

The zonal load data is given for each of the eight ERCOT weather zones. I mapped counties in ERCOT to weather zones by hand. I combine this data with two products from the U.S. Energy Information Administration (EIA): data on electricity delivered to customers in 2020 (EIA Form 861 Part C, Delivery Companies) and utility service territories, which reports the counties served by each utility that delivered electricity in ERCOT in 2020 (Form 861, Service Territories).³⁰ The EIA data reports, for the calendar year of 2020, the total delivered electricity by sector (residential, commercial, industrial, and transportation) for each transmission and distribution utility that operated in ERCOT. I allocate those deliveries to counties based on a two-step weighting procedure. First, I look at a county's share of households (per the American Community Survey), and commercial and industrial employment (per the Census Bureau's County Business Patterns) within a utility's service area.³¹ This yields the share of a utility's deliveries that are estimated to belong to each utility-county-sector pair. Then, I generate county weights based on each county's total utility deliveries across utilities. As a result, the EIA data determine the spatial allocation of load within weather zones, while the ERCOT data determine the level of load for each hour in the data.

²⁹Curtailment in the Eastern weather zone in 2020 is 12.7%, which comes from a single solar resource that entered the grid in November of 2020.

³⁰County load allocations are robust to using different years of EIA Form 861 data. Form 861 Part C begins in 2020. 6 of 214 counties have no coverage based on EIA data. All six of these counties are in the North weather zone, which has 44 counties; I assign all six 1/44 of North weather zone load and renormalize the other 38 county weights such that all weights add to one.

³¹EIA also reports a small (<0.1%) share of electricity delivered for transportation purposes, which I allocate to counties based on their commercial employment share.

Table 5: Production Summary Statistics by ERCOT Weather Zone

Variable	2018	2019	2020	2021	2022	2023	2024
<i>Overall</i>							
Average price (\$/MWh)	37.9	46.2	24.2	169.5	72.6	62.7	29.8
Average generation (GW)	43.0	43.8	43.3	44.7	49.0	51.0	52.9
Average demand (GW)	43.0	43.8	43.4	44.8	49.2	51.0	52.9
Renewable curtailment (%)	2.3	2.4	4.3	5.1	4.9	4.2	4.7
<i>Coast</i>							
Average price (\$/MWh)	37.8	45.1	24.3	129.1	78.7	62.6	29.0
Average generation (GW)	9.4	9.9	9.5	10.2	10.2	10.7	11.3
Average demand (GW)	12.0	12.1	12.2	12.6	13.1	13.6	13.8
Renewable curtailment (%)	–	0.0	0.1	2.4	1.2	1.4	2.6
<i>East</i>							
Average price (\$/MWh)	37.3	45.0	24.3	202.2	77.3	65.6	29.3
Average generation (GW)	4.9	4.8	4.8	4.6	4.9	4.9	4.3
Average demand (GW)	1.7	1.6	1.6	1.7	1.8	1.8	1.8
Renewable curtailment (%)	–	–	12.7	1.3	0.5	1.0	2.5
<i>Far West</i>							
Average price (\$/MWh)	35.8	53.1	33.6	83.9	61.4	56.7	33.1
Average generation (GW)	2.7	3.1	3.6	3.5	4.2	4.5	4.8
Average demand (GW)	2.9	3.4	3.7	3.9	4.7	5.8	6.8
Renewable curtailment (%)	3.3	3.7	5.1	5.5	6.7	5.4	4.9
<i>North</i>							
Average price (\$/MWh)	35.5	40.6	18.1	172.0	66.7	59.4	23.5
Average generation (GW)	5.1	5.2	5.3	5.0	5.7	5.6	6.0
Average demand (GW)	1.0	1.0	0.9	1.1	1.3	1.4	1.7
Renewable curtailment (%)	2.5	2.1	4.3	5.7	2.7	3.2	4.4
<i>North Central</i>							
Average price (\$/MWh)	39.2	46.3	23.4	203.7	75.2	67.7	29.2
Average generation (GW)	9.6	9.6	8.8	9.1	10.6	11.7	12.2
Average demand (GW)	13.7	13.6	13.2	13.4	14.7	14.5	14.8
Renewable curtailment (%)	1.2	1.7	2.9	3.8	3.1	2.3	2.3
<i>South</i>							
Average price (\$/MWh)	36.9	43.7	21.4	155.5	54.0	41.7	26.6
Average generation (GW)	3.6	3.8	3.8	4.1	4.6	4.6	4.7
Average demand (GW)	3.5	3.6	3.5	3.6	3.9	4.0	4.2
Renewable curtailment (%)	1.0	1.5	4.1	6.2	6.7	5.9	7.2
<i>South Central</i>							
Average price (\$/MWh)	37.4	47.8	24.2	196.0	75.9	68.5	32.4
Average generation (GW)	5.4	4.9	4.6	4.7	5.3	5.3	5.7
Average demand (GW)	6.9	7.0	6.9	7.2	8.2	8.4	8.5
Renewable curtailment (%)	0.1	0.0	0.0	0.5	0.9	1.9	1.7
<i>West</i>							
Average price (\$/MWh)	34.7	40.4	20.0	176.1	66.5	62.4	31.3
Average generation (GW)	2.4	2.6	3.0	3.5	3.5	3.6	3.9
Average demand (GW)	1.4	1.4	1.4	1.4	1.5	1.4	1.3
Renewable curtailment (%)	2.6	2.5	4.1	3.8	5.5	4.7	6.0

Notes: Prices are hourly county LMPs weighted by estimated county demand. County LMP coverage begins July 1, 2018, so the 2018 price averages cover July–December; the remaining statistics cover the full calendar year. Generation is the SCED Base Point integrated over five-minute intervals. Overall generation includes all SCED resources. Renewable curtailment is $\max\{\text{HDL} - \text{Base Point}, 0\}$ for wind and solar divided by renewable generation plus curtailment. Regional rows are ERCOT weather zones.

226 substations appeared in the congestion data for congested elements on bulk transmission lines. Linking ERCOT’s congestion data to physical grid segments required a manual review of ERCOT substation names. I used ERCOT’s Major Transmission Element (MTE) datasets to map ERCOT’s abbreviated station names to full station names. There were several situations where substation names did not appear in the MTE data; in these cases, I performed review of other ERCOT and public reports to map the abbreviations to full substation names. I then matched full substation names to data from OpenStreetMap, a public infrastructure database that contains geolocations for electrical infrastructure worldwide. OpenStreetMap is frequently used in energy research (see, e.g., xxx, yyy, zzz). OpenStreetMap data provided precise locations for correctly mapped substations. For substations that did not have a corresponding entry in OpenStreetMap, I searched documentary evidence to determine the county that they were located in. I was able to successfully find county information for 215 of 226 total substations in the congestion data.

ERCOT also has a number of Generic Transmission Constraints (GTCs). GTCs monitor a collection of elements and control flows across all of them simultaneously. I identified 26 GTCs in the congestion data, all of which I was able to construct constituent transmission lines or geographic monitoring regions for.³²

³²Region-based GTCs monitor all of the flows into and out of regions. None of the region-based GTCs were used in the model of the grid described in Section XXX because they did not bind on bulk transmission lines sufficiently often.

B Prices and Quantities Under Arbitrary Grid Configurations

Entry to the electric grid is a function of expected prices and expected quantities. Prices and quantities, in turn, depend on the dispatch process: where electricity comes from and where it travels to. This section details the steps I take to estimate dispatch under any potential configuration of generation on the grid.

I describe this process in five steps. First, I give a brief overview of the network representation I use and my estimation of power flows.³³ Next, I describe the dispatch problem in more detail and how it maps to price formation at the nodal level. I then describe the two estimation tasks I have: estimating unobserved grid capacity and modeling the process by which the grid operator commits dispatchable units to produce. Finally, I discuss validation of the model I build.

B.1 Network

The representation of the grid that I use in this article treats counties as nodes connected by edges. Edges are based on the bulk transmission grid; when any two neighboring counties have at least one 230 kV or 345 kV transmission line connecting them, they are given an edge on the graph representation I use. In cases where multiple transmission lines connect two counties, I aggregate those transmission lines into a single edge.

ERCOT uses AC transmission. This means that grid operators cannot control where power travels on the network. Instead, grid operators turn generators on and allow demand to take power off the grid at places where its needed. This lack of control means that grid operators must be careful to not turn generators on in such a way that the flow of power will overload any transmission line on the grid. I now describe the process by which grid operators measure power flows on the network.

Power flows depend on net injections and a transmission line’s reactance (i.e., how strongly a line limits power flows) to power. Injections matter based on direction and size: nodes that inject electricity to the grid tend to push power towards nodes that withdraw electricity from the grid; larger injections push more power away from a node and larger withdrawals pull more power towards a node. Reactance depends on edge length and edge voltage. Longer edges have higher reactances, while higher voltage edges have lower reactances.

³³Note that my power flow estimation is intended to recover economically relevant congestion exposure and renewable revenues and makes assumptions that would be inappropriate for a full engineering model of the grid.

I assign each underlying transmission line in the grid a reactance rate of 0.32 ohms/km.³⁴ I calculate reactance for edge c_{ij} (which links nodes i and j) as a per km measure based on length L_c and voltage V_c :

$$X_c = \frac{0.32 \times L_c \times 100}{V_c^2} \quad (19)$$

Note that reactance rates are common across all edges. Assigning common reactances omits details about equipment like conductor material and transformers from my model. Further, because the grid connections I consider are almost entirely 345 kV lines, almost all heterogeneity in line reactance comes from line lengths. [Will add cites throughout here, this is all standard electrical engineering to my knowledge]

Power flows on edge c_{ij} can now be solved. I use a DC power-flow model to do solve for flows (DC power-flow models are common in industry; ERCOT uses a DC power-flow model to solve dispatch, see XXX). I make several assumptions in order to estimate the DC power-flow model: I ignore resistance and line losses, which are less concerning at shorter distances (relative to, e.g., longer distance transmission considered by Hausman (2024)); and I assume net injections sum to zero (so the grid is balanced) and phase angle differences are small (so there are not blackouts and equipment still works). Allowing θ_i to represent a node's phase angle (which measures the net injection of node i relative to the rest of the network), power flows F_c on edge c_{ij} are equal to:

$$F_c = \frac{\theta_i - \theta_j}{X_c}. \quad (20)$$

Solving the grid operator's problem requires understanding how an injection at node i travels throughout the grid. To do so, I construct a nodal susceptance matrix, where edge susceptance is the inverse of edge reactance: $b_{ij} = \frac{1}{X_c}$. Let B be the nodal susceptance matrix, entries of which give the susceptance of all edges on the grid: $B_{ij} = -b_{ij}$. For unconnected nodes, entries are set to 0. Diagonal entries are the sum of susceptances for all edges connected to a node: $B_{ii} = \sum_{j:(i,j) \in \mathcal{E}} b_{ij}$, where $\mathcal{E}$ contains all edges connected to node i . Then, nodal balance is satisfied if $p = B\theta$, where p is the vector of county net injections and θ is the vector of all phase angles (insert cite).

Phase angles are unidentified in levels; instead, they are defined in relation to a reference county (called a slack county in the literature). I set Harris county as the slack county and modify the B matrix by removing the Harris row and column, yielding the reduced matrix

³⁴0.32 is standard because of XXXX. Some edges represent multiple transmission lines. I weight underlying edges in proportion to their length and calculate a weighted average reactance across all constituent lines for multiple line edges.

$\tilde{B}$.³⁵ The phase angle of each non-Harris county is then given by the following:

$$\tilde{\theta} = \tilde{B}^{-1}\tilde{p}. \quad (21)$$

$\tilde{B}^{-1}$ is the inverse of the reduced B matrix, $\tilde{p}$ is the vector of non-Harris net injections, and $\tilde{\theta}$ is the corresponding vector of phase angles.

Finally, we can calculate Power Transfer Distribution Factors $PTDF_{c,r}$, which describe how a one-MW injection in r changes power flows on edge c_{ij} (conditional on a one-MW withdrawal in Harris county). The PTDF for corridor $c_{i,j}$ from an injection in region r is given by the following:

$$PTDF_{c,r} = \frac{1}{X_c}[(\tilde{B}^{-1})_{i,r} - (\tilde{B}^{-1})_{j,r}] \quad (22)$$

Note that injections in region r can have effects on power flows far from the injection site. c does not have to touch r . PTDFs are bound by $[-1, 1]$, where negative numbers indicate power flowing in the opposite direction from a randomly chosen reference direction for a given corridor.³⁶ Collecting all PTDFs in a matrix $PTDF$ shows how a one-MW injection in any region changes power flows throughout the grid.

The $PTDF$ matrix can be used to calculate flows on corridors across the grid. Given a vector of net injections p , the flows on edge c are given by:

$$F_c = \sum_r PTDF_{c,r} p_r. \quad (23)$$

Note that this construction of flows, which is what I use in estimation, does not rely on the assumption I made about reactance.

B.2 Grid Capacity

After reducing the network to the county-edge graph, I have 300 total edges for which I need capacity information. I estimate K_c in four parts. First, I set initial capacities for each edge in the graph. Initial capacities are based on direct limits taken from the shadow price data when available. The shadow price data only provides limits for transmission lines that are actively constrained. Of the 204 nodes, 50 receive estimated limits based on the shadow price data. The remaining 250 edges receive capacities based on their voltage and the number of

³⁵The reference-node choice is an accounting normalization: for any balanced vector of net injections, predicted physical flows are invariant to the choice of slack node.

³⁶A PTDF is bound by $[-1, 1]$ because the injection is one MW.

lines aggregated within a single edge.

Voltage measures the pressure that transmission exerts on power to flow in a given direction – not the maximum amount of electricity that flows along a given transmission line. The two are proportional however. I assume that all remaining 345 kV edges have a capacity of 1200 MW, which is the average binding limit for 345 kV transmission lines based on the congestion price data. For consistency, I assume that 230 kV transmission lines have a capacity of 800 MW.³⁷

In the second step of the process, I use evidence from observed dispatch. Based on observed net injections in a county – i.e., looking at, for each period from 2018-2024, observed generation and estimated demand – PTDFs allow me to calculate the implied flows on a given edge as follows:

$$F_{ct} = \sum_r PTDF_{cr}(q_{rt} - D_{rt}). \quad (24)$$

If an edge has direct evidence of an implied power flow greater than its initial capacity and it did not cause congestion during period t , I adjust the capacity of that line upwards to fit that data.

In the third step of the process, I solve the grid operator’s dispatch problem for each hour from 2018-2024. To do so, I reconstruct historical supply stacks using generator offers from the SCED data and observed dispatch limits.³⁸ Then I solve dispatch based on the current best guess of edge capacities.

I compare the solved dispatch model to observed outputs at the edge level. For each edge, I look at the spread between the model implied and observed nodal endpoint prices. I calculate 90th percentile spreads; if the model spread is too high, I increase edge capacity,

³⁷A 1,200 MW real-power limit on a balanced three-phase 345 kV AC circuit corresponds to a line current of about 2.0 kA:

$$I = \frac{1200 \text{ MW}}{\sqrt{3} \times 345 \text{ kV}} \approx 2.01 \text{ kA},$$

where I is line current. I interpret this value as a short-corridor thermal-capacity prior rather than a long-distance transfer capability. This distinction is consistent with engineering loadability guidance: AEP reports that 345 kV lines have much lower long-distance loadability under the surge-impedance-loading yardstick, but that short-distance loadability can be higher when conductor and station-equipment thermal ratings are adequate; it gives 1,500 MW on a 345 kV line as approximately 3.8 SIL over about 50 miles (American Electric Power (2008); Dunlop et al. (1979)). Consistent with NERC facility-rating standards, observed facility ratings depend on equipment ratings, ambient assumptions, operating limits, and verified engineering analysis (North American Electric Reliability Corporation). There are 25 edges with voltage classes of 230 kV. Assuming these edges have a capacity of 800 MW is consistent for a 230 kV line with a line current of about 2.0 kA. In reality, capacity depends on many factors, including conductor size, the number of conductors, weather, substation equipment, line length, and more.

³⁸I also account for committed units required output.

and if it's too low, then I decrease edge capacity.³⁹ Then I look at the modeled vs observed binding frequency for each edge. I define observed binding as anytime the edge's absolute endpoint price spread is greater than \$5 MWh.⁴⁰

This process results in small adjustments to modeled capacities. It increases flows for 3% of edges and decreases flows for 17% of edges. The maximum decrease is 356 MW, the 10th percentile decrease is 26 MW, and the 25th percentile decrease is 0 MW. Similarly, the maximum increase is 391 MW, while the 99th percentile increase is 61 MW and the 95th percentile increase is 0 MW. It improves the prices calculated by the dispatch model relative to the observed data by roughly 10%.

[to come: brief description of how I handle the WESTEX GTC. deep appendix material.]

B.3 Commitment

This section describes the steps I take to account for unit commitment in the dispatch process. Unit commitment places a (sometimes large) stack of price insensitive dispatchable supply at the bottom of the offer stack. This shifts the offer curve for generators bidding into the market based on marginal cost to the right and depresses prices as a result. In order to have an accurate counterfactual model of prices and quantities, I must account for the grid operator's commitment behavior in equilibrium.

A model of endogenous commitment is outside the scope of this paper. Instead, I estimate a reduced form approximation of commitment each period. Estimating commitment requires two steps. First, I estimate the MW capacity of dispatchable resources available to commit each hour. I estimate availability separately for coal and natural gas. The model depends on unit online statuses. Online statuses reflect whether or not a generator is currently producing electricity. Online dispatchable resources must produce some electricity (or else they pay ramping costs to turn off). They are able to produce up to a High Sustained Limit (HSL). I model the expected amount of available coal as equal to the sum of HSLs for all online coal units in a given hour.⁴¹ I model the expected amount of available natural gas as the sum of HSL across online units, plus α times the HSL of all offline gas units. I find that setting $\alpha = 0.40$ produces the best fit with prices. Implicitly, this suggests that as dispatch conditions shift, 40% of offline gas is available for commitment.

The second step of the commitment model estimates actual commitment. To estimate

³⁹I look at 90th percentile spreads because differences are only informative when constraints are binding. Maximum spreads measure prices during blackouts, which are difficult to accurately price.

⁴⁰This part of the process is not that important but the way I'm doing this is pretty ugly, so worth reconsidering for the next time I run the model.

⁴¹This assumes that counterfactual entry would not change whether or not units were online in any given period.

the model, I predict commitment separately by county and fuel based on local and system net load, renewable production, and system tightness, which measures how close system net load is to the total HSL of all online and offline coal and natural gas units. I include fixed effects for hour, month, and market-period.⁴² The regression produces, for each counterfactual hour, dispatchable availability, which I apply to submitted coal and natural gas offer curves.

⁴²I include five market periods in the model: pre-Winter Storm Uri, from January 1, 2018 to February 12, 2021; Uri, from February 13 to 19, 2021; Post-Uri, from February 20 to June 7, 2021; 2021 Reform, from June 8 to December 31, 2021; and Post-reform, from January 1, 2022 onwards. ERCOT adopted a more conservative operating posture starting in summer of 2021, committing resources more actively. In 2022, ERCOT made changes to its Operating Research Demand Curve, which determines how the units that ERCOT forcibly commits get compensated for that commitment.